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Parametric LCA of a Piggott Wind Turbine

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Glossary

ADP(MR) Abiotic Depletion Potentiel, Material Ressources. 1, 27–30, 34, 35

AEP Annual Energy Production. 6, 7

AFPM Axial Flux Permanent Magnet. 4, 16

EF Ecological Footprint. 23

EoL End of Life. 2, 6, 10, 35

FU Functional Unit. 6, 7

GWP Global Warming Potential. 1, 27–30, 34

LCA Life Cycle Assessment. 1–4, 6, 9, 10, 12, 20–22, 34

LCI Life Cycle Inventory. 2, 3, 12

PEF Product Environmental Footprint. 1, 23

SWT Small Wind Turbine. 1–3

Abstract

This study explores the parametric environmental assessment and optimization of locally manufactured Small Wind Turbines (SWTs). The studied system is a horizontal-axis wind turbine designed and built by the community of self-builders. The goal of this work is to support design decisions based on environmental performance by identifying trade-offs between key geometric parameters: mast height and rotor diameter.

To achieve this, we developed a Python tool that integrates `OpenAFPM` for generator modeling, `Brightway 2` for Life Cycle Assessment (LCA), and `NoLoad` for non-linear optimization. A parametric inventory was built using the `lca.algebraic` library, enabling flexible modeling of system components as functions of design variables.

The environmental impacts were assessed using the Product Environmental Footprint (PEF) methodology. Results of the sensitivity analysis show that Abiotic Depletion Potential, Material Resources (ADP(MR)) is mainly driven by rotor diameter due to copper usage, while Global Warming Potential (GWP) is primarily influenced by mast height due to the presence of steel and concrete. Afterwards, a constrained optimization was performed to minimize the environmental score per kWh produced under different site conditions, using a logarithmic wind profile to model wind speed as a function of height.

The analysis reveals that optimal configurations vary depending on site characteristics and that a trade-off exists between maximizing energy production and minimizing environmental impact. Thus, while the rotor diameter consistently reaches its upper bound, indicating no internal optimum, the mast height exhibits an optimal value that balances material usage with energy gains. Therefore, a large rotor on a short mast seems to be preferable from an environmental point of view rather than the opposite. This work demonstrates the value of combining parametric LCA and optimization to inform environmentally conscious design of low-tech energy systems.

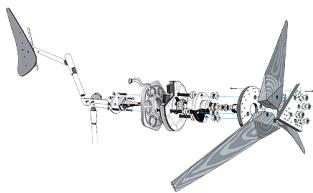
1.1 Presentation of the laboratory

This internship is co-supervised by two laboratories. It is financed by the laboratory Ampère, a laboratory based in the city of Lyon in France, but I was staying in the G2E Lab in Grenoble. The Ampère Laboratory operates under the supervision of the following institutions: INSA Lyon, CNRS, Claude Bernard University Lyon 1, and École Centrale de Lyon. It was formed through the merger of the Lyon Electrical Engineering Center (CEGELY) and the Lyon Industrial Automation Laboratory (LAI), and is distributed across three sites, one in Écully and two in Villeurbanne. The laboratory is organized into three departments: the Electrical Energy (EE) department, which focuses on electricity production, transmission, and conversion; the Automation for Systems Engineering (AIS) department, which studies system optimization and dynamics; and the Bio-Engineering department, which develops methods and applications in bio-engineering. One of my supervisors, Hugo Helbling, belongs to the department EE and another, Vincent Léchappé, to AIS. The last one, Adrien Prévost, is a postdoctoral fellow at the G2Elab laboratory.

1.2 The Piggott wind turbine

This internship is focused on the Piggott wind turbine, a small locally manufactured wind turbine. Originally developed in the 1980s by the Scottish inventor Hugh Piggott, the turbine was created to provide a simple, low-cost renewable energy solution for off-grid communities. The Piggott turbine is characterized by its hand-crafted construction and the use of relatively simple materials such as steel, copper, wood, and resin. The blades are traditionally carved from wood, the stator consists of copper coils cast in resin, and the rotors are equipped with permanent magnets mounted on steel discs.

What makes the Piggott wind turbine unique is its unpatented design, which has enabled its global diffusion and adaptation. Hugh Piggott has published a detailed construction manual *A Wind Turbine Recipe Book* [1], accessible in multiple languages, which has made it possible for individuals and communities around the world to build their own turbines. In France, the Tripalium network, a collective dedicated to promoting small-scale wind turbines, has produced its own handbook [2], based on the translation of the work of Hugh Piggott. The network also organizes workshops across the country where participants build functional turbines while learning the principles of wind energy and craft skills.



(a) Exploded view



(b) Wind turbine in operation

Figure 1.1: Photos of the Piggott wind turbine, from [3]

1.3 Context of the study

This internship fits into a larger research effort to evaluate and improve the sustainability of small-scale wind energy systems. It continues the work initiated by Adrien Prévost during his PhD at the laboratory Ampère, which suggests improvement for the Piggott wind turbine in several respects [4]. Firstly, he shows that an active rectifier could strongly improve the performance of the wind turbine. He has also conducted Life Cycle Assessments (LCA) which highlight the battery's and structure's impact on the ecological balance sheet. Finally, he has proposed a parametric model of the structure in order to find a height which minimizes the ratio between impacts and electricity production.

The present study aims to continue to explore sustainable design approaches for small wind turbines, with a specific focus on how LCA can be used as an evaluation and decision-making tool. The goal is to carry on the framework proposed by [5] by including the wind generator in the parametrization. It is also to pay particular attention to the End of Life (EoL) of the system and to highlighting impact transfers.

1.4 State of the art and problematic

While the urgency of global warming is pushing more and more players to deploy renewable energies, other issues, such as pollution and the availability of resources, are becoming increasingly important. So there is an urgent need to find energy sources that are sustainable in all these respects. The question of measuring environmental impact is therefore becoming crucial in order to determine the most sustainable systems.

LCA can be used to fulfill this objective. LCA is a standardized method for assessing all the environmental impacts of a product or a service over its lifetime. This includes the extraction of the materials, the fabrication, the use, and the EoL. It is defined by the two ISO standards, 14040 and 14044 [6,7]. According to these standards, LCA consists of four main phases, which are shown in Figure 1.2:

- Goal and scope definition, where system boundaries and the functional unit are established.
- Life Cycle Inventory (LCI), involving data collection of all material and energy flows.
- Life Cycle Impact Assessment where these flows are translated into environmental impact categories (e.g., climate change, resource depletion).
- Interpretation, which informs decisions by analyzing results in light of uncertainties and assumptions.

Since the early 2000s, several studies have applied LCA to assess the environmental footprint of wind turbines [8] [9]. Even if most of these studies are carried out on large wind turbines, several have been conducted on small ones [10]. On the methodological side, the author of the reference [4] explains that all the studies reviewed in the paper use energy production as a functional unit. The phase cradle-to-gate, i.e., extraction, fabrication and transport, is covered in all studies. However, the EoL has been poorly researched. Only three studies in seven take it into account, and they all underline a lack of information.

For the conclusions, there is a consensus on the fact that small wind turbines (SWT) have far less impact than diesel generators [10]. In addition to that, all studies agree on the fact that results are strongly linked to wind resources, as it is a key parameter in energy production.

Some research efforts were carried to investigate which scale of system is the most effective in terms of energy footprint (environmental impacts / kWh). On the one hand, [9] concludes that

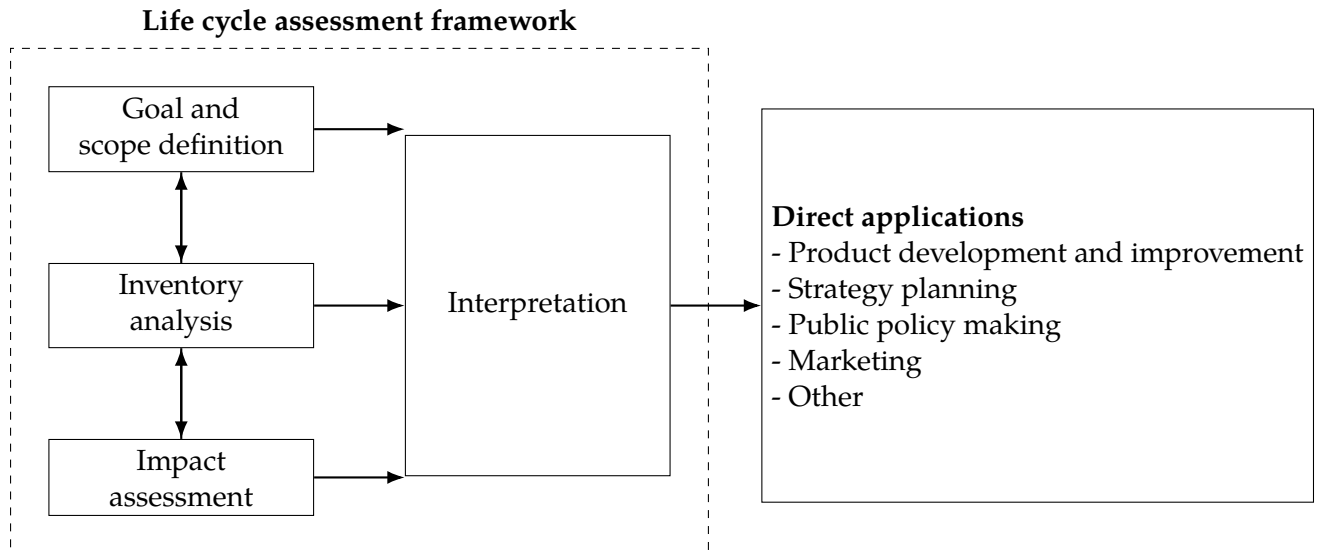


Figure 1.2: Life Cycle Assessment (LCA) framework with its main components and applications

a large wind turbine (3 MW) is more eco-efficient than a small one (250 W). Yet, the SWT studied here is a vertical axis model, which is known to have a poorer performance than horizontal axis designs [11]. On the other hand, [12] explains that “across GWP100 it is that the larger turbines have a higher burden compared to that of the smaller turbines.” Finally, [8] finds no influence of size on the footprint.

In this state-of-the-art review, only one paper has studied the trade-off between hub height and impacts [5]. It concludes that increasing the height of the mast is, in most cases, a good way to reduce impact per unit function and that the optimal height mainly depends on the roughness surface length of the terrain and the local wind resource. However, increasing the height of the mast increases the absolute impact of the system what should not be overlooked.

In most studies, LCA models are static, because they are assessing the impacts of existing systems. Parametric LCA is an emerging field, hence the small number of studies. [5] implemented the parametrization on the size of the mast and of the batteries but doesn’t scale the turbine itself. [13] proposes a framework for parametrized LCA in four steps, which as been adapted to our current study :

1. **Preliminary Information:** This stage involves collecting all the data needed for the inventory. Thanks to the fact that the Piggott wind turbine is not patented, information gathering is simpler than for other systems. Here, we will be relying mainly on the Tripalium self-build manual mentioned above.
2. **Development of a Life Cycle Inventory (LCI)** Implementation of the data collected and binding to background process.
3. **Identification of Key Parameters:** They are divided in two categories. The background parameters, which depend on society’s infrastructure system i.e energy mix. And the foreground parameters which depend on the design of the product. For our study, we choose two foreground parameters which are the one chosen by the person building the wind turbine, i.e. the height of the mast and the rotor diameter. We will also put a constraint on the quantity of energy produce.
4. **PLCA Model Construction:** Consist in modelling each life cycle phase using foreground parameters as independent variables. Background parameters are treated as constants. The total environmental impact is computed as a function of these parameters.

After this review, several gaps can be identified :

- Most LCAs on small wind systems are static and fail to explore parameter variations such as mast height or rotor diameter.
- The End-of-Life phase is frequently oversimplified or excluded altogether.
- The impact transfer between impacts categories is not properly studied which can lead to counter-productive solution
- There is a lack of open-source, reproducible LCA scalable models for small wind turbine.
- Existing tools often overlook user-configurable design trade-offs.

This internship project addresses the current gaps in the literature by developing a practical and flexible LCA tool for the Piggott wind turbine. This work is built on one main objective :

This study aims to explore design trade-offs between mast height and rotor diameter, in order to support the locally manufactured wind turbine community in selecting the most sustainable system configuration based on specific installation site conditions.

With this main objective come side ones :

- **Develop a parametrized LCA tool** in Python, connected to the technical outputs of OpenAFPM, an open-source software used to model Axial Flux Permanent Magnet (AFPM) generators.
- **Investigate the End-of-Life** phase to understand how each components are treated
- **Highlight impact transfers**

In this report, I will first present the methodology used to established the LCA. Then, I will give details about the Python tools I have created for this LCA. Finally, I will present the main results on the trade-off between height and diameter. The study will be focused on designs coming from the Tripalium handbook [2].

2.1 Goals and scope definition

As previously mentioned, the primary objective of this study is to investigate the trade-off between the increase of energy production, by increase of mast height and turbine diameter, and the reduction of environmental impacts.

The methodological framework applied here is adapted from the one developed in [4]. The system which has been under studied was a typical energy production site featuring a Piggott wind turbine for off-grid or on-grid configuration. This setup is referred to as the *technical system*. Its purpose is to meet the energy demands of a *sociosystem*, defined as a system of interacting human agents. Both the technical system and the sociosystem are components of a broader system, the *anthroposystem*, which itself operates within the *ecosphere*. These relationships and interactions are illustrated in Figure 2.1.

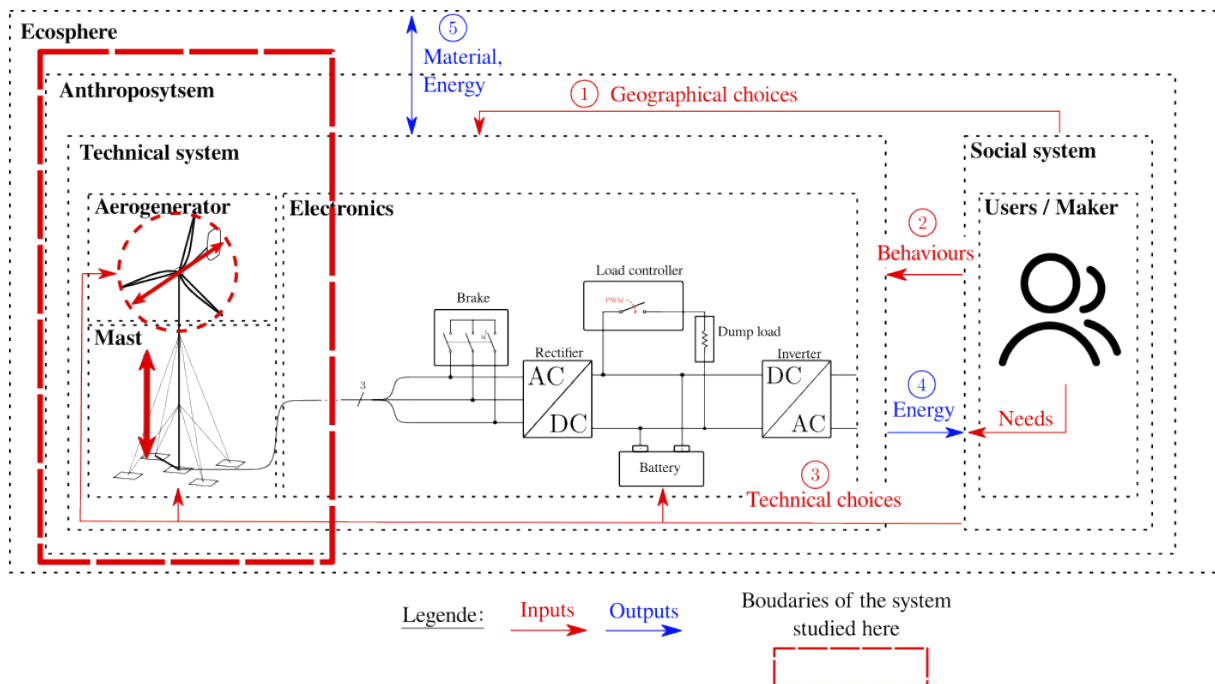


Figure 2.1: Inital and actual systems, adapted from [4]

Users or designers make decisions about the location (1), consumption (2), and specific technical aspects (3) of the technical system, which determine its configuration. The system then provides energy (4) to the sociosystem and exchanges materials and energy with the ecosphere (5). The environmental impacts of the technical system can be evaluated based on these exchanges with the ecosphere.

From this study framework, we define a narrower one, which stops at the output of the alternator, i.e. the wind turbine and the mast, as it represents on Figure 2.1. The output of our system is the energy at the alternator output. This study aims at investigating technical trade-offs for the design of the wind turbine, so it is why the framework focuses on the electro-mechanical part. Nevertheless, the electronic and electrical parts, especially the batteries, can represent more than the half of the impacts so it would require a dedicated work in order to take it into account properly. That was not the purpose of this study. It should be noted that the modelling carried out here can be linked to a modelling of the electronic part.

The levers studied are :

- geographical location, implying different wind resource and surface roughness length
- choice of rotor diameter and mast height
- energy requirements

In this study, only the **extraction, fabrication and use phases are assessed**. The EoL has been investigated but the data collected in the field has not yet been implemented. This will be the subject of a future study.

2.2 Functional unit

Regardless of the objective, quantifying the environmental impacts of an activity requires a precise definition of what that activity produces and the function it fulfils. This is where the concept of the Functional Unit (FU) becomes essential: it is a quantitative metric that reflects both the function and the amount of output associated with the activity.

In this work, we use as a FU :

The total electricity production in kWh at the alternator output per day over 20 years on the European continent.

This approach deviates from classical LCA methodology, which assumes a fixed functional unit for comparability. Here, it will depend on the design of the wind turbine and the site location. This deviation is justified by the prospective nature of the study, which aims to identify optimal system configurations rather than evaluate a predefined one. The energy need is treated here as a variable to be optimized.

Since the system configurations (diameter and mast height) vary, it is necessary to express the FU as a function of these parameters. The goal of this section is to describe how this parametric FU is computed.

Annual Energy Production Data

To begin with, we use estimations of the Annual Energy Production (AEP) in kWh for different wind turbine diameters and average wind speeds. These values are sourced from Table 1, page 9 of the Tripalium handbook [2]. The AEP data is structured as shown in Table 2.1, with wind speed (in m/s) as the index and turbine diameters (in mm) as columns.

	Diameter [mm]				
Windspeed [m/s]	1800	2400	3000	3600	4200
3	144	264	408	588	804
4	360	648	1020	1464	1992
5	636	1116	1752	2520	3432
6	888	1572	2460	3552	4824

Table 2.1: Annual Energy Production (AEP) in kWh for different wind speeds and turbine diameters.

To normalize energy output across different turbine sizes, we compute the AEP per swept area (in kWh/m²). The swept area $A(d)$ of a turbine of diameter d (in mm) is given by:

$$A(d) = \frac{\pi}{4} \left(\frac{d}{1000} \right)^2 \quad (2.1)$$

We then average these values across diameters to obtain the mean glsaep per swept area as a function of wind speed.

Regression of AEP per Swept Area

A second-degree polynomial regression is performed on the averaged values of AEP per swept area versus wind speed. Figure 2.2 shows that energy production per unit area is nearly independent of turbine diameter. The regression has a coefficient of determination R^2 of approximately 0.999, indicating an excellent fit.

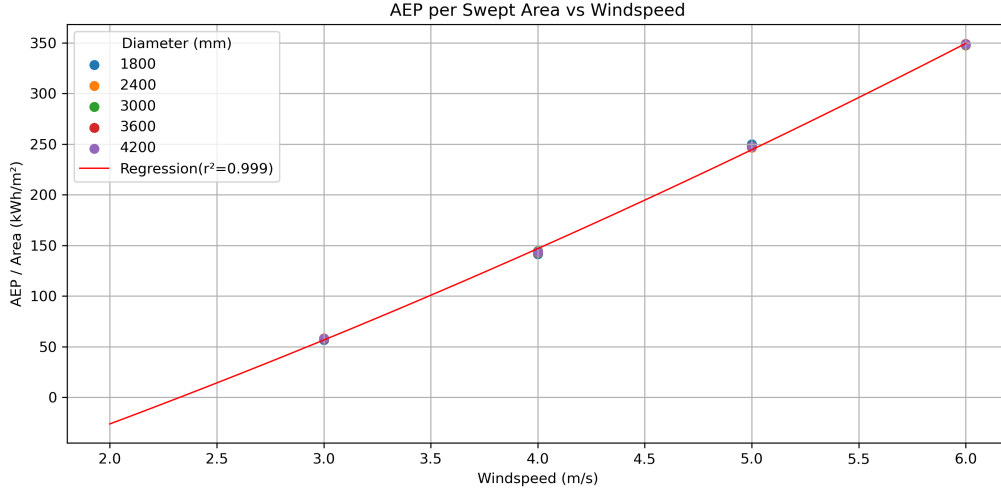


Figure 2.2: AEP per swept area (kWh/m²) versus average wind speed. Polynomial regression is shown in red.

Final FU Expression

To compute the energy produced over the turbine's lifetime, we account for site-specific wind speed variation as a function of mast height. The average wind speed at height h is estimated using a logarithmic wind profile, which accounts for surface roughness through the parameter z_0 , known as the *roughness length*. This value characterizes the effect of terrain on wind flow: low values (e.g., $z_0 = 0.01$ m) correspond to smooth surfaces like open water or short grass, while higher values (e.g., $z_0 = 0.5$ m) correspond to rougher terrain such as forests or urban areas.

Given a reference wind speed v_{ref} (in m/s) measured at height h_{ref} (in m), the wind speed $v(h)$ at mast height h is estimated by:

$$v(h) = v_{\text{ref}} \cdot \frac{\ln(h/z_0)}{\ln(h_{\text{ref}}/z_0)}$$

It is important to note that the turbulences are not taken into account here, but they can have a significant impact on production.

Finally, the functional unit $FU(d, h)$, representing the total energy produced over 20 years, is:

$$FU(d, h) = \text{AEP}_{\text{reg}}(v(h)) \cdot A(d) \cdot T_{\text{lifetime}} \quad (2.2)$$

where $\text{AEP}_{\text{reg}}(v(h))$ is the output of the polynomial regression evaluated at the estimated average wind speed.

2.3 Life Cycle Inventory

2.3.1 Inventory overview

The life cycle inventory was carried out using the *Brightway2* software with the *Ecoinvent 3.8 cut-off* database. It is primarily based on the inventory published in [5]. Most of the input data come from the Tripalium handbook [2]. Volumes were calculated from the dimensions provided in the handbook, and then converted into masses by multiplying by the material densities.

The inventory is organized into two distinct systems, each further divided into subsystems:

- **The structure**, which includes the foundation, mast, and rigging.
- **The wind generator**, which consists of the rotor (propeller), alternator, and furling system.

The Figure 2.3 shows this different subsystems.

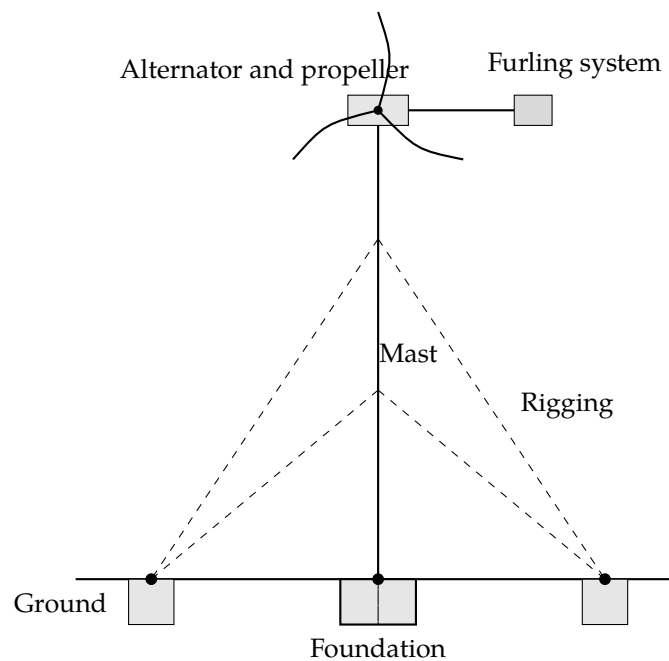


Figure 2.3: Simplified schematic of a Piggott wind turbine.

In order to explore optimal configurations, the inventory had to be parameterized. Further details are provided in the next section, but the overall approach involves performing regressions on designs presented in the handbook, which includes masts of 12, 18, or 24 meters in height, and wind turbines with diameters of 1.80, 2.40, 3.00, 3.60, or 4.20 meters. The parameterization of the mast has already been established in previous work [4]; thus, the main contribution of this study is the parameterization of the wind turbine as a function of its diameter.

Some background activities have also been modified in order to improve the accuracy of the life cycle inventory. This following subsection provides an overview of the main changes made to the data and modeling choices.

2.3.2 Activities

This is only a part of the total inventory. You can find the complete one in Appendix 1.

Market and Transport

In previous work, production activities were preferred over market datasets in order to make transport processes explicitly visible in the inventory. This approach is retained in the present study. However, due to time constraints, transport distances have not yet been parameterized. Since transportation contributes only a small share of the total environmental impact, this omission is not expected to significantly affect the results.

Concrete

Concrete is used for the turbine's foundation. The original model used the activity *concrete block production, RoW*, which, upon further investigation, was found to represent breeze blocks. As wind turbine foundations are poured rather than block-built, this has been replaced with *concrete, all types to generic market for concrete, normal strength, RoW*, which better reflects a generic poured concrete formulation. Also, concrete has been scaled in this LCI which was not the case in [5].

Copper

Copper is used in two main components: the cable linking the alternator to the rectifier, and the stator windings. Both were initially modeled using the activity *market for copper, cathode, GLO*. However, in order to assure insulation, cables in the stator are enameled. The activity and associated quantities for enamel coating were derived from the LCA of an electric traction motor [14].

Steel

Two grades of steel are used to represent the main structural components: *unalloyed* and *low-alloyed* steel. In this inventory, only hardware (such as bolts and nuts) is modeled with low-alloyed steel. The remainder, primarily composed of S235 structural steel, is represented using the unalloyed steel activity.

Zinc Coating

In earlier work, the galvanization of guy wires was modeled using the activity *zinc coating, coils, RER*, which corresponds to continuous coating of long, flat steel products like sheet metal rolls. This does not accurately reflect the production method of guy wires, which are twisted first and then galvanized. Therefore, this has been replaced by *zinc coating, pieces, RER*, applied to the external surface area of the assembled guy wires.

	Sub-system	Element	Element in french	Part	Process (Ecoinvent 3.8)	Lifetime/End of life
Wind generator	Propeller	Blades	Pales	Dry soft wood	planing, beam, softwood, u=20%-Europe without Switzerland	With proper annual maintenance and protective tape on the leading edge, the rotor blade is a wear part but can last up to 20 years. A common issue is damage caused by the tail vane or leading-edge erosion due to wind, rain, or hail. Repairs are possible using epoxy resin, which provides good durability. At end of life, the blade can be incinerated or disposed of with wood waste.
		Blades attachment	Fixation des pales	Plywood	plywood production-RER	
	Stator	Copper coil	Bobine de cuivre	Copper, cathod	market for copper, cathode-GLO	Copper recovery is sometimes accepted by metal recyclers, although not always. At end of life, the resin can be burned or melted, allowing copper to be recovered. Without protective paint, the resin degrades under UV exposure and rain, leaving the fiberglass exposed—resulting in a lifespan of around 5 years. If protected, the coating should be renewed every 5 to 10 years, enabling a service life of 20 years or more. Main failure risks include short circuits, impacts, or excessive friction. In most cases, the stator is not recovered, as it tends to break more easily than other components.
				Sheet rolling	sheet rolling, copper-RER	
				Wire drawing	wire drawing, copper-RER	
		Enamelling	Revêtement isolant	Resin	polyester resin production, unsaturated,RER	
				Xylene	xylene production,RER	
				Electricity	market group for electricity, medium voltage,RER	
		Resin	Résine	Polyester resin	market for polyester resin, unsaturated	
				Talc	steatite quarry operation-RoW	
				Glass fibre	glass fibre production-RER	

Figure 2.4: Extract from the inventory

2.4 Lifetime and end of life

Handling end-of-life scenarios is always a key issue of LCA. This study is no exception. The lack of information make the implementation of the EoL challenging. To address this issue, information was collected directly from the field using two approaches.

The first source of information is a week-long self-build course that I have participated to, organized by Ti'éole [15], a company that is part of the Tripalium network and that is specialized in the deployment of small wind turbine.

The second, is an in-depth interview with Jay Hundall, founder and director of Ti'éole. The interview support was based on the LCI shown in Figure 2.4. We went through each item and J.Hundall provided details of the lifespan he estimated, the end-of-life management and the main causes of failures.

Main Findings on Wind Turbine Components and Lifespan

The feedback from practitioners provided several key insights regarding the durability, maintenance, and recyclability of components:

- **Expected lifespan:** When well constructed and regularly maintained, the turbine can operate at least 20 years. It could more but the oldest Piggott wind turbines in France have 19 years so we don't have any more hindsight.
- **Construction and build quality:** Service life is highly dependent on construction quality. While basic assembly during a self-build course takes about one week, an additional week is often needed to ensure proper finishing.
- **Maintenance:** Proper maintenance is essential to long-term operation. Annual maintenance involves visual inspection, tightening fasteners, re-greasing, and replacing wear parts such as blade tape. Every five years, a full disassembly is recommended to check key components. If all goes well, only wearing parts need to be changed.

- **Foundations:** Concrete foundations are typically left in place due to the difficulty of removal. However, it can be reused for other installations, for example if the wind turbine is replaced.
- **Reuse :** A lot of elements of the wind turbine can be reused, such as rotor, mast or propeller. Stators are fragile and harder to recover after failure.
- **Recyclability:** Most components are recyclable via standard scrap channels. The stator, which consists of copper coils cast in resin, is not always accepted by scrap merchants. Wood parts can be incinerated or treated as wood waste.

The complete result for lifespan and end of life are available at the Appendix 1.

2.5 Methodology Summary and Key Assumptions

This study focused on evaluating the environmental impacts of locally manufactured small wind turbine, the Piggott wind turbine. The goal is to assess trade-offs between height and diameter, to minimize impact per energy production, rather than to model complete real-world installations. The scope and assumptions are structured to support this goal.

Key Assumptions

- **Production-focused:** The study models only electricity production; electricity consumption and demand-side variability are not included.
- **System boundaries:** The system includes the wind turbine up to the rectifier. Downstream components such as batteries and grid connection are excluded.
- **End-of-life:** End-of-life treatment is not implemented in the inventory model. Recovery and recycling potentials are discussed qualitatively.
- **Component lifespan:** A default lifetime of 20 years is assumed for nearly all components, based on the field data collection.

3.1 General Interactions Between Software

Several tools and software packages are used in this study to perform the life cycle assessment (LCA).

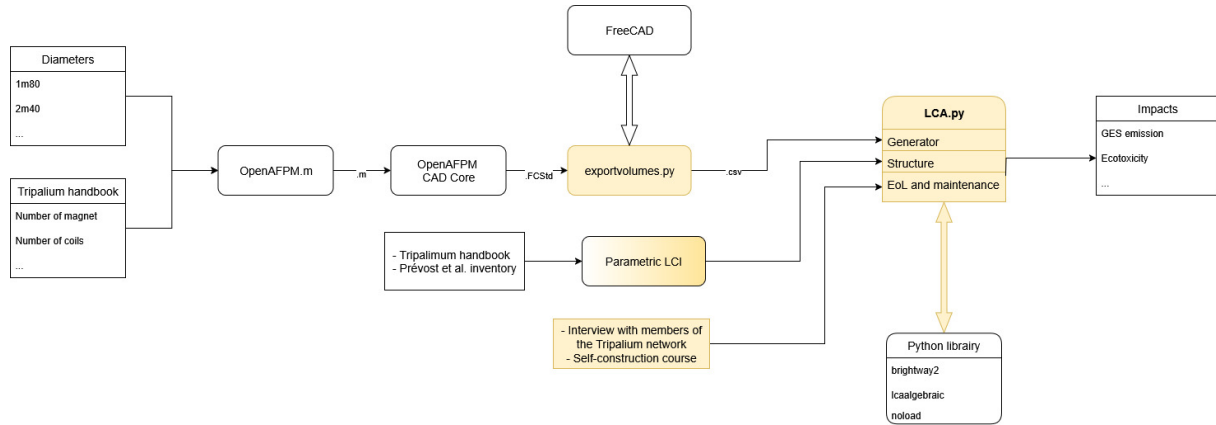


Figure 3.1: Interactions between all tools and software used in this work. In yellow, the elements added by this study.

The general approach involves creating regressions for the mass of each component as a function of diameter and height. These regressions are then implemented in the life cycle inventory (LCI) to parametrize it and compute environmental impacts.

The LCI and LCA are carried out using *Brightway2* and the Python library *lca.algebraic*, which is “a layer above Brightway2 designed for the definition of parametric inventories with fast computation of LCA impacts¹.”

Subsequently, the *NoLoad* Python library is used for optimization [17].

We now examine each stage in detail to explain how the life cycle analysis was conducted. All input data are sourced from the Tripalium handbook [2].

3.2 Sizing of the Structure

3.2.1 Foundation

In [5], because the diameter of the wind turbine was fixed, the volume of concrete was also considered as a constant value. However, in practice, it depends on both the height of the mast and the diameter of the wind turbine.

Height [m]	Diameter [mm]				
	1800	2400	3000	3600	4200
12	1.5	2.0	2.5	2.5	3.5
18	—	2.0	2.5	2.5	3.5
24	—	—	3.5	3.5	5.0

Table 3.1: Concrete volume [m³] required for each combination of mast height and turbine diameter. Source: Tables 1 and 2, p.69 [2].

¹Citation from [16]

Based on this data, we create a regression using the following bi-variable quadratic form:

$$f(h, d) = a_0 + a_1h + a_2d + a_3h^2 + a_4hd + a_5d^2$$

where:

- h is the mast height,
- d is the rotor diameter,
- a_i are the regression coefficients.

The results are shown in Figure 3.2.

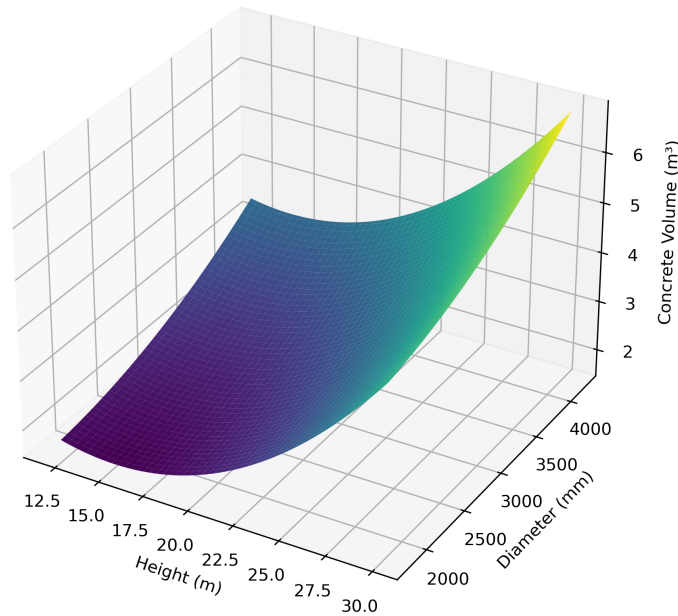


Figure 3.2: Fitted concrete volume model.

3.2.2 Steel Tower

The tower is constructed from steel pipes, with component dimensions varying according to mast height. The structure comprises three main tubular elements:

- **Mast**
- **Yaw pipe:** A secondary pipe attached near the mast base to assist with raising and lowering the turbine.
- **Pivot pipe:** The pipe that connects the mast to the wind turbine nacelle.

The mast and yaw pipe are composed of 6-meter sections. Additional dimensions are shown in Table 3.2.

Mast Height [m]	Mast		Yaw		Pivot		
	D [mm]	e [mm]	D [mm]	e [mm]	D [mm]	e [mm]	L [mm]
12	88.9	3.2	60.3	2.9	60.3	5.0	395
18	114.3	3.2	88.9	3.2	76.0	5.0	480
24	114.3	3.2	88.9	3.2	76.0	5.0	480

Table 3.2: Dimensions of tower components (diameter D , thickness e , and length L for the pivot) as a function of mast height. Source: Table 7, p.110 [2].

From this data, we generate a regression, shown in Figure 3.3.

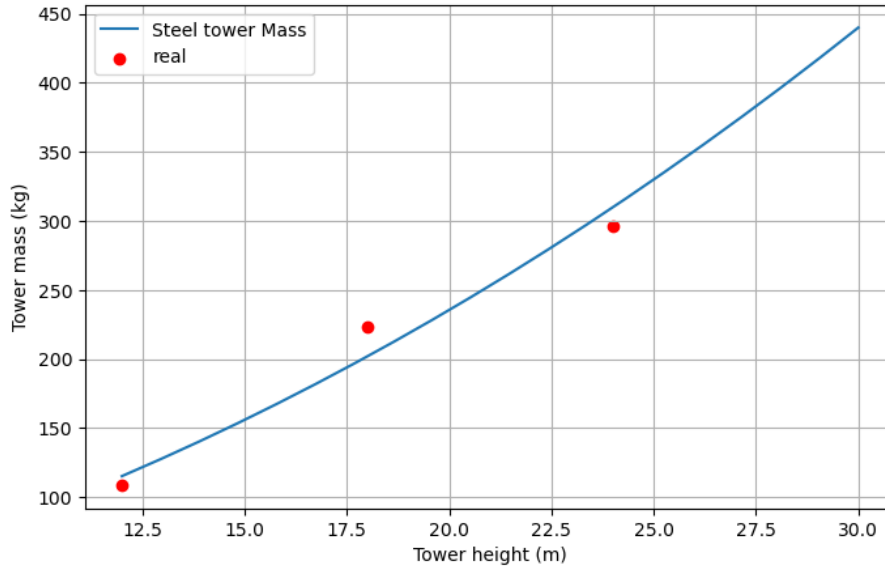


Figure 3.3: Mass of the steel tower as a function of mast height. Regression in blue, actual values in red.

3.2.3 Rigging

Guy Wires

Guy wires are used to keep the mast upright. They are arranged in layers of four, with one layer fixed at the top and additional layers added every 6 meters.

The length of the top guy wires L_c is given by:

$$L_c = \frac{4h}{\sin(\alpha)} = \frac{8h}{\sqrt{3}}$$

where:

- $\alpha = 60^\circ$ is the angle between the cable and the ground,
- h is the mast height.

Smaller cables are used to reinforce the mast, with a constant diameter of 6 mm. These are connected to the same anchor points as the top cables and to the mast at heights of 6 m, 12 m, and 18 m, with lengths λ_1 , λ_2 , and λ_3 :

$$\lambda_1 = 4\sqrt{\left(\frac{h}{2}\right)^2 + 6^2} \quad \lambda_2 = 4\sqrt{\left(\frac{h}{2}\right)^2 + 12^2} \quad \lambda_3 = 4\sqrt{\left(\frac{h}{2}\right)^2 + 18^2}$$

We then define two functions: the “real” stepped function corresponding to discrete layers of stays, and a smoothed regression without breakpoints (Figure 3.4). The latter is used to facilitate optimization.

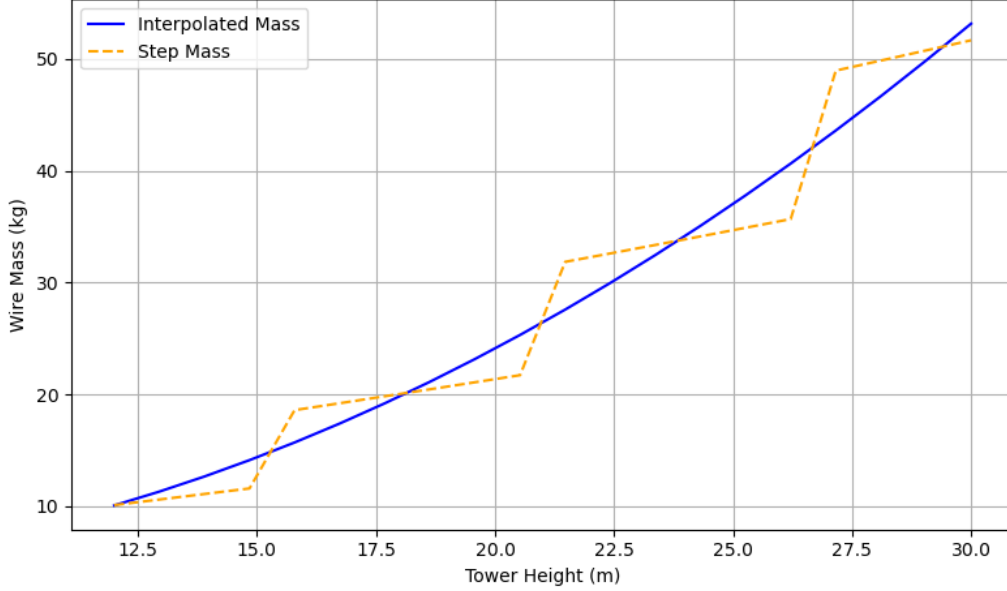


Figure 3.4: Mass of the guy wires as a function of mast height. Regression in blue, stepped model in yellow.

To compute the area coated with zinc, we multiply the external perimeter of the cable by the total cable length:

$$A_{\text{tot}}(h) = \pi D L_{\text{tot}}(h)$$

where:

- A_{tot} is the total coated surface area,
- D is the diameter of the guy wire,
- $L_{\text{tot}}(h)$ is the total length of guy wires.

Hardware

All screws, nuts, cable clamps, and other fastening components required for structural assembly are included in the model. The majority of the weight is attributed to the rigging hardware. All components are galvanized.

To estimate the mass of each component, we referred to the Tripalium manual and cross-checked the specifications with manufacturer data. Full details, including hardware references and masses, are provided in Appendix 2. A summary example is shown in Table 3.3.

Mast Height [m]	Hardware Examples [kg]		...	Total Hardware Mass [kg]
	Expansion Bolts	Turnbuckle Fork		
12	0.144	5.05	...	12.2
18	0.348	6.65	...	18.1
24	0.348	9.05	...	23.2

Table 3.3: Hardware mass by mast height. Example hardware includes expansion bolts and turnbuckle forks.

3.3 Sizing of the wind generator

This part corresponds to the branch connected to the generator cell in Figure 3.1.

3.3.1 Creation of a 3D Model

To compute the mass of each component of the wind generator, a 3D model is created to extract all relevant volumes.

OpenAFPM

To begin with, we use OpenAFPM [18], a modeling tool designed for the parametric design of AFPM generators in wind energy systems. This open-source tool, available online, requires input parameters such as rotor radius, number of poles, and other design characteristics (see Figure 3.5(a)). It then computes the electromagnetic properties of the generator (Figure 3.5(b)) and outputs a .mat file containing all calculated data.

Create a MagnAFPM Simulation

Title * WT_2m4

Blade Rotor Inputs	Generator Stator Inputs	Generator Rotor Inputs
Turbine radius * 1.2 m	Select coil type * Rectangular	AFPM rotor topology * Double Rotor
Aerodynamic coefficient * 0.3	Winding fill factor * User defined	Magnetic material * NSFeB N40
Cut-in wind speed * 3 m/s	Winding fill factor * 0.55	Number of generator poles * 12
Nominal wind speed * 10 m/s	Stator thickness * User defined	Magnet length * 45 mm
Cut-in TSR * 8.75	Stator thickness * 13 mm	Magnet width * 30 mm
Design TSR * 7	Current density * 6 A/mm ²	Magnet thickness * 10 mm

(a) Inputs

Generator Rotor Outputs

Property	Value	Variable name
Rotor disk thickness	10 mm	hr
Magnet width-to-pole pitch ratio	0.348969	ai
Inner-to-outer magnet radius	0.80985	kd
Flux density in airgap	0.577701 T	Bmax
Flux density on magnet surface	0.645413 T	Bmg
First harmonic of flux density in airgap	0.374032 T	Bp
Flux per pole (Calculated)	0.00089364 Wb	Flux_pole
Magnets on a rotor disk	16	magnet_num
Distance between magnets	46.9355 mm	dist_magnet_Rin
Magnet mass	3312 g	MagnetMass
Magnet density	7.5 g/cm ³	mag_density

(b) Outputs

Figure 3.5: User interface of the OpenAFPM software

As shown in Figure 3.1, the design parameters used as inputs in OpenAFPM are derived from the Tripalium handbook [2]. All generated .mat files are available in the directory LCA/eco-optimization-tool/SizingWT/OutputsOpenAFPM.

OpenAFPM CAD

Following this step, the OpenAFPM CAD tool [19] is used to generate a 3D CAD model from the OpenAFPM outputs. The generator inputs (referred to as MagnAFPM) are loaded directly from the .mat file, while additional data concerning the furling system and user-defined parameters must be entered manually.

MagnAFPM

Rotor Diameter	Rotor Topology	Rotor Disk Radius	Rotor Disk Inner Radius
2400	Double	150	103.25
Rotor Disk Thickness	Magnet Length	Magnet Width	Magnet Thickness
10	40	30	10
Magnet Material	Number Magnet	Stator Thickness	Coil Type
Neodymium	12	13	1
Coil Leg Width	Coil Hole Width at Outer Radius	Coil Hole Width at Inner Radius	Mechanical Clearance
21.5	30	30	3
Inner Distance Between Magnets	Number of Coils per Phase	Wire Weight	Wire Diameter
20	3	2.6	1.4

(a) Inputs for the generator

Furling

Vertical Plane Angle	Horizontal Plane Angle	Bracket Length	Bracket Width
20	55	300	30
Bracket Thickness	Boom Length	Boom Pipe Diameter	Boom Pipe Thickness
5	1000	48.3	5
Vane Thickness	Vane Length	Vane Width	Offset
6	1200	500	125

User

Blade Width	Hub Pitch Circle Diameter	Rotor Disk Central Hole Diameter	Holes Diameter
124	100	85	12
Metal Length L	Metal Thickness L	Flat Metal Thickness	Yaw Pipe Diameter
20	6	10	20.3

Visualize

(b) Inputs for the propeller and the furling system

Figure 3.6: User interface of OpenAFPM CAD

To ensure consistency, a table was created to match each input with corresponding data from the Tripalium handbook. An excerpt of this input mapping table is shown in Figure 3.7; the full table is provided in Appendix 3.

Parameter	Values					Reference (Tripalium handbook "Construire une éolienne", 2023)
	1m80	2m40	3m	3m60	4m20	
Furling						
Vertical Plane Angle	20	20	20	20	20	p33
Horizontal Plane Angle	55	55	55	55	55	p33
Bracket Length	300	300	400	500	600	Table 10 p36
Bracket Width	30	30	30	30	30	Table 10 p37
Bracket Thickness	3	3	3	3	3	Table 10 p38
Boom Length	800	1000	1200	1500	1800	Table 9 p35
Boom Pipe Diameter	48,3	48,3	48,3	48,3	48,3	Table 9 p35
Boom Pipe Thickness	3,2	3,2	3,2	3,2	3,2	Standard thicness for this diameter http://
Vane Thickness	6	10	10	12	12	Table 10 p36
Vane Length	1000	1200	1300	1700	2000	Table 10 p36
Vane Width	510	635	890	890	1145	Table 10 p36
Offset	100	125	150	200	250	Table 4 p30 Table 6 p32

Figure 3.7: Mapping between OpenAFPM CAD input fields and data from the Tripalium handbook

Finally, a complete 3D model of the wind turbine is generated using OpenAFPM CAD (see Figure 3.8). The resulting file is saved in the .FCStd format used by FreeCAD. All 3D files corresponding to the various designs in the handbook are available in the directory LCA/eco-optimization-tool/SizingWT/WT-3D-Files.

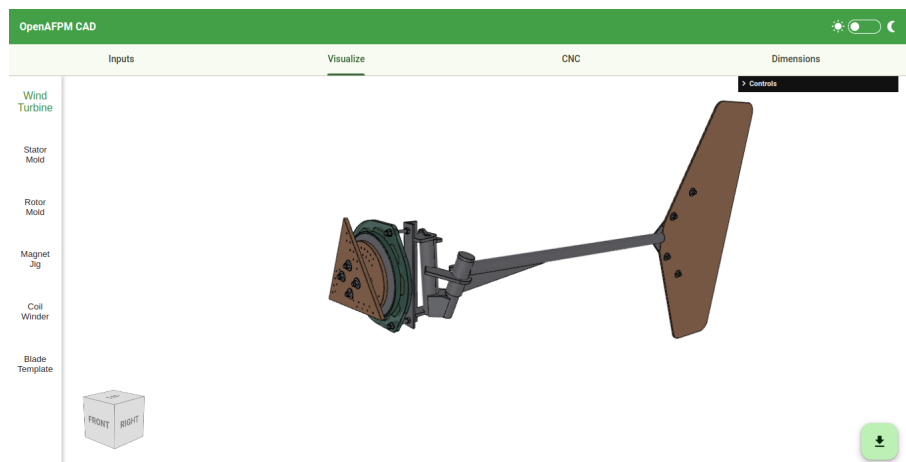


Figure 3.8: 3D model of a Piggott wind turbine generated with OpenAFPM CAD

3.3.2 Mass Computation

A key feature of FreeCAD is its compatibility with Python, allowing Python scripts to be executed as commands. Leveraging this capability, we developed a script named *export-volumes.py*, which automatically traverses each FreeCAD project file and extracts the component volumes into a CSV file.

Instructions for using this script are provided in the README file located in the directory LCA/eco-optimization-tool/SizingWT.

After executing *export-volumes.py* on each .FCStd file, we obtain output tables such as the one shown in Table 3.4, where the volume of each subcomponent is listed.

Component	Volume [cm ³]
WindTurbine	38745.66
WindTurbine/Alternator	14186.10
WindTurbine/Alternator/Stator	3868.90
WindTurbine/Alternator/Stator/Coils	1234.96
WindTurbine/Alternator/Stator/ResinCast	2633.93
WindTurbine/Alternator/Rotor_Back	2749.41
WindTurbine/Alternator/Rotor_Back/Rotor_Disk_Back	1536.97
WindTurbine/Alternator/Rotor_Back/MagnetResin_Assembly	1212.44
WindTurbine/Alternator/Rotor_Back/MagnetResin_Assembly/Rotor_ResinCast	992.74
WindTurbine/Alternator/Rotor_Back/MagnetResin_Assembly/Rotor_Magnets	219.70
WindTurbine/Alternator/Rotor_Front	2746.93
WindTurbine/Alternator/Rotor_Front/Rotor_Disk	1534.49
WindTurbine/Alternator/Rotor_Front/Rotor_Disk/BaseFeature	1536.97
WindTurbine/Alternator/Rotor_Front/Rotor_Disk/Binder	1536.97
WindTurbine/Alternator/Rotor_Front/MagnetResin_Assembly	1212.44
WindTurbine/Alternator/Hub	559.74
WindTurbine/Alternator/Hub/Flange	362.53
WindTurbine/Alternator/Hub/Flange_Cover_Front	30.19
WindTurbine/Alternator/Hub/Flange_Cover_Back	40.84
WindTurbine/Alternator/Hub/StubAxleShaft	126.18
WindTurbine/Alternator/Studs_Hub	295.69

Table 3.4: Excerpt from the volumes-4200.csv file, generated by the export-volumes.py script

Then, all the .csv files are merged into a single DataFrame, a tabular data structure used in Python. The components are then grouped into relevant categories. After converting volumes to masses where necessary (by applying appropriate material densities), a complete inventory of the wind turbine is obtained for each of the five rotor diameters, as shown in Table 3.5.

Component	Diameter [mm]					Unit
	1800	2400	3000	3600	4200	
Blades	10.7	14.1	37.2	69.4	80.6	kg
BladesAttachment	5.65×10^{-4}	1.25×10^{-3}	2.45×10^{-3}	2.90×10^{-3}	3.09×10^{-3}	m ³
CopperCoils	2.3	2.63	4.25	4.65	6.09	kg
ResinStator	1.5	1.78	2.44	2.97	3.58	kg
MetalDisks	7.14	10.5	14.7	19.6	24.1	kg
SurfaceDisk	0.606	0.671	0.937	0.998	1.23	m ²
Magnet	0.824	2.47	2.47	3.30	3.30	kg
ResinRotor	0.559	1.69	2.28	2.40	2.70	kg
Hub	3.98×10^{-4}	3.98×10^{-4}	3.98×10^{-4}	5.51×10^{-4}	5.60×10^{-4}	m ³
Vane	2.43×10^{-3}	6.01×10^{-3}	9.10×10^{-3}	1.41×10^{-2}	2.13×10^{-2}	m ³
BoomPipe	2.85	3.56	4.27	5.34	6.41	kg
HingeOuter	1.47	1.70	1.94	2.22	2.43	kg
Bracket+Support+TailStop	1.86	2.41	3.12	4.83	6.15	kg
Frame	3.30	3.77	4.46	5.35	5.94	kg
YawPipe	1.63	1.95	2.29	2.61	2.87	kg
YawPlates	0.404	0.549	0.655	3.80	5.05	kg
Hardware inox	2.48	2.59	2.66	4.59	4.80	kg
Hardware zinc	2.67	2.67	2.67	2.72	2.72	kg
MouldStator	3.04×10^{-3}	4.54×10^{-3}	5.40×10^{-3}	6.34×10^{-3}	7.35×10^{-3}	m ³
MouldRotor	4.53×10^{-3}	6.56×10^{-3}	1.03×10^{-2}	1.03×10^{-2}	1.48×10^{-2}	m ³

Table 3.5: Inventory of the wind generator for five diameters.

3.3.3 Regressions

Finally, quadratic regressions are computed for each component to parametrize the inventory as a function of rotor diameter. As an illustration, the regression for the volume of the vane plate is shown in Figure 3.9.

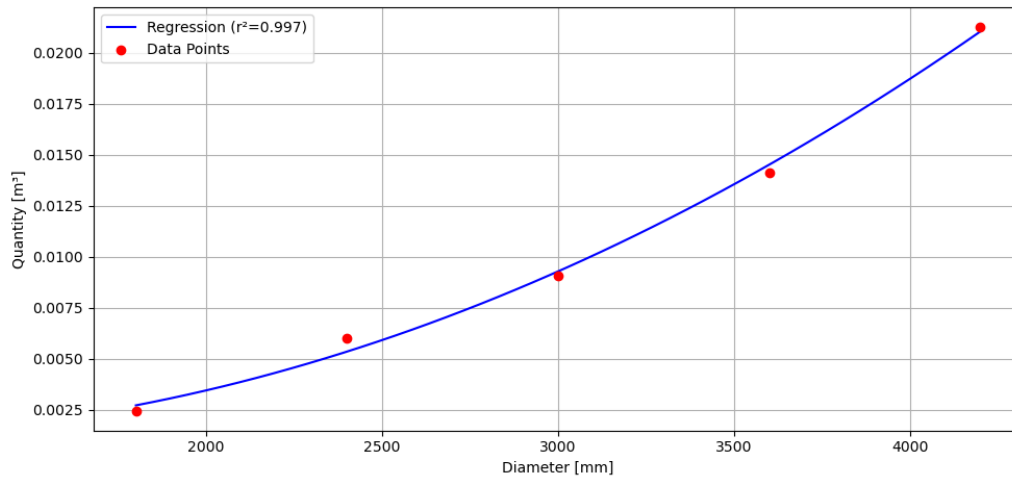


Figure 3.9: Volume of the vane plate in m³ as a function of diameter in mm

3.4 Implementation in the code

Now that a parametric inventory has been established as a function of both mast height and rotor diameter, we can examine the structure of the corresponding code implementation.

3.4.1 Structuration with `lca.algebraic`

System Modeling

As previously mentioned, `lca.algebraic` is a Python library that serves as an additional layer over Brightway 2. It provides a range of classes and additional functionality that enable the creation of parametric life cycle inventories and which facilitate impact assessment as well as sensitivity analysis.

This modular structure allows users to build a complete system activity by nesting simpler activities, following a hierarchical approach similar to Russian dolls. Figure 3.10 illustrates the hierarchical structure of the complete system, with a specific focus on the foundation branch. Background processes sourced from ecoinvent are shown in *italics*.

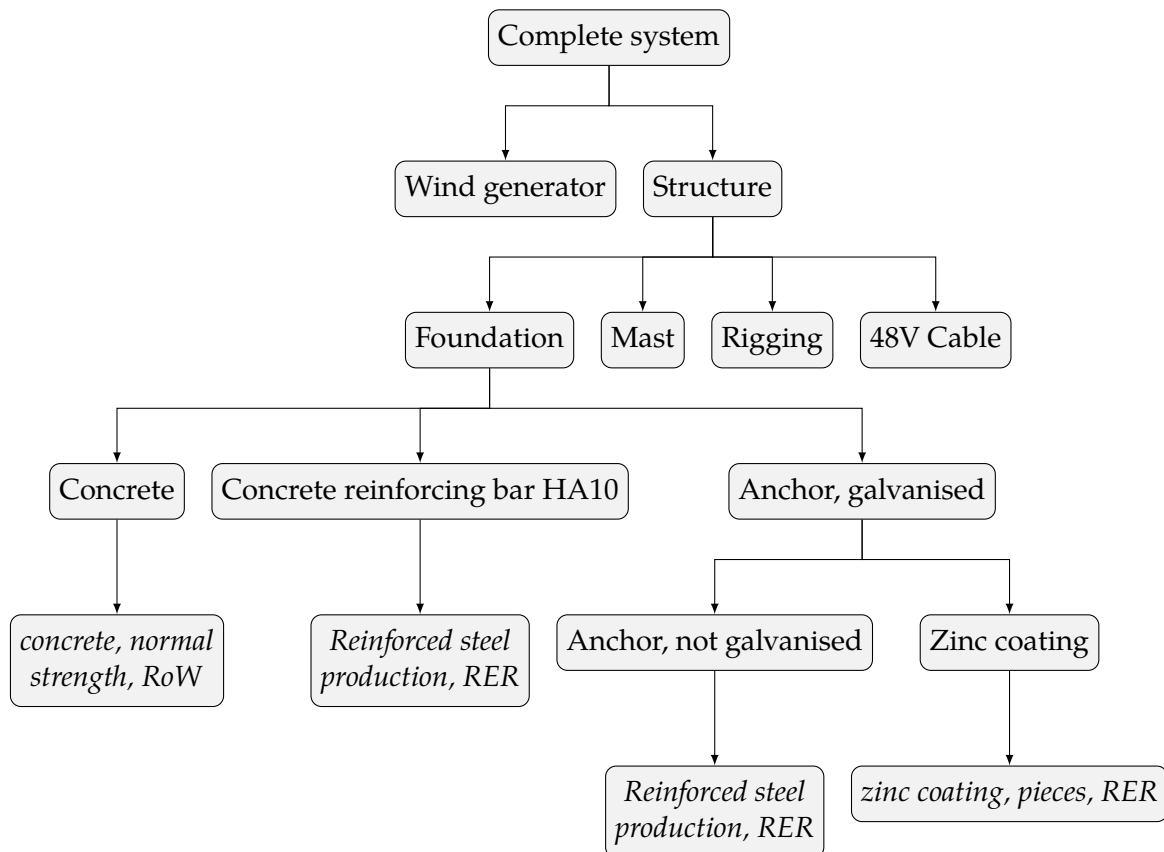


Figure 3.10: Tree structure of the complete system for LCA modeling, with a focus on the foundation branch. Background processes from ecoinvent are shown in *italics*.

Functionality for sensitivity analysis

One the key features of `lca.algebraic` is the possibility to label any activity with an "axis". An axis is a metadata that can group the activities into sub-categories. It can be any type of grouping, material, phase of the life cycle, element of the system etc. It allows to plot the impacts according to this sub-categories, which can be very useful to perform sensitivity analysis. Axis for the branch foundation are shown Figure 3.11

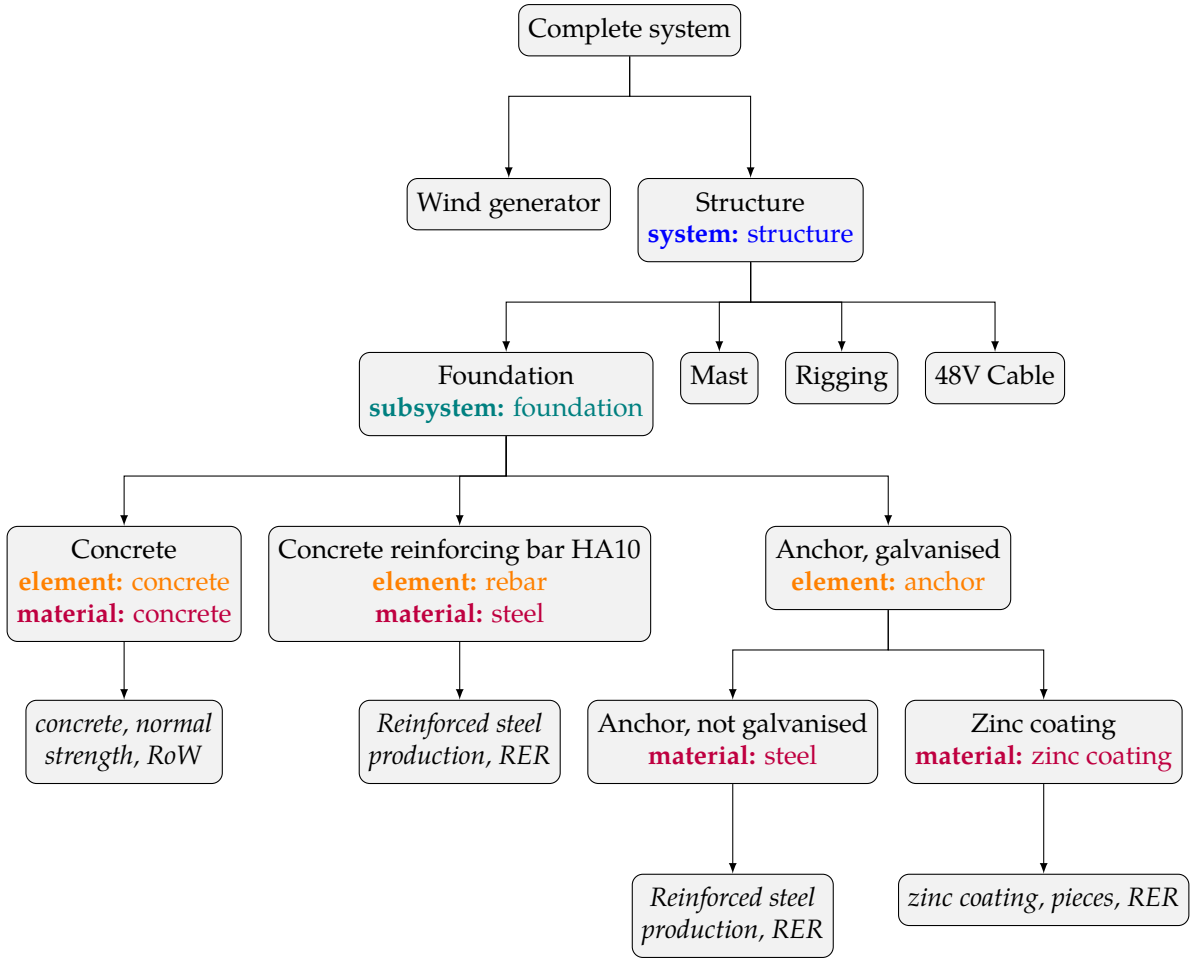


Figure 3.11: Tree structure of the complete system for LCA modeling, with a focus on the foundation branch. Axis tags are color-coded: **system**, **subsystem**, **element**, **material**.

3.4.2 Optimization with NoLoad

NoLoad (Non-Linear Optimization using Automatic Differentiation) is a Python library developed by G2ELab that provides tools for solving optimization problems, “it aims to be a lightweight non-linear optimization tool for energy components and systems².”

In this project, the stochastic optimization method is used. Although this approach is significantly slower than other methods offered by NoLoad, such as the least squares algorithm, it is currently the only compatible option. This limitation arises because Brightway 2 does not support the data structures Array Box used by NoLoad to propagate derivatives, making derivative-based methods like least squares unusable.

There are discussions to re-implement Brightway 2’s matrix operations in a way that allows for derivative propagation. However, at present, only the stochastic method is supported.

3.4.3 Calculation of uncertainty

One of the major strengths of Brightway 2 is its ability to perform advanced uncertainty propagation using the Monte Carlo method. However, this method is not compatible with parametrized databases, as it interprets them as containing empty amounts, as illustrated in Table 3.6.

An attempt was made to export the parametrized database generated with `lca.algebraic`,

²Citation from [17]

evaluate each formula for a given height and diameter, and then re-import the evaluated database. However, the re-importation process failed. Further investigation is required to resolve this issue.

Amount	Activity	Location	Database	Formula
0.0 kg	Copper coil	GLO	bdd_syst	$0.96 \cdot d \cdot (1.9391 \times 10^{-7} \cdot d + 0.0004349) + 0.7646$

Table 3.6: Exemple of an entry in the database of the system, with a formula and a empty amount.

3.5 Conclusion

This chapter presented the development of a parametric LCA for small wind turbines by integrating several open-source tools: OpenAFPM for generator modelling, FreeCAD for 3D design, Brightway2 for LCA, lca.algebraic for parametric inventory structuring and No Load for optimization. Together, these tools enabled the creation of a highly adaptable inventory capable of reflecting changes in design variables like mast height and rotor diameter.

This part was particularly challenging. We tried many different ways before arriving at this one. It is not fully automatic as it required user intervention to create the 3D files but it is a compromised between the final objective and the available time. The main difficulties was the compatibility issues between all the different tools and python libraries. It took a long time to get everything working together, and this was made possible by the fact that all them are open source.

4.1 Impacts assessments

To assess the environmental impacts of the system, we use methodology known as the Product Environmental Footprint (PEF), which is recommended by the European Commission [20]. This methodology is based on the Environmental Footprint (Ecological Footprint (EF)) method, which includes sixteen impact categories, shown in Figure 4.1. In this study, we use version EF v3.0.

The sixteen impact indicators are normalized and weighted to produce an aggregated single score. This approach provides a standardized treatment of scalar quantities, which facilitates eco-optimization and enables comparison across scientific studies. However, it also involves applying weighting factors that, although defined by the European Commission, are open to debate.

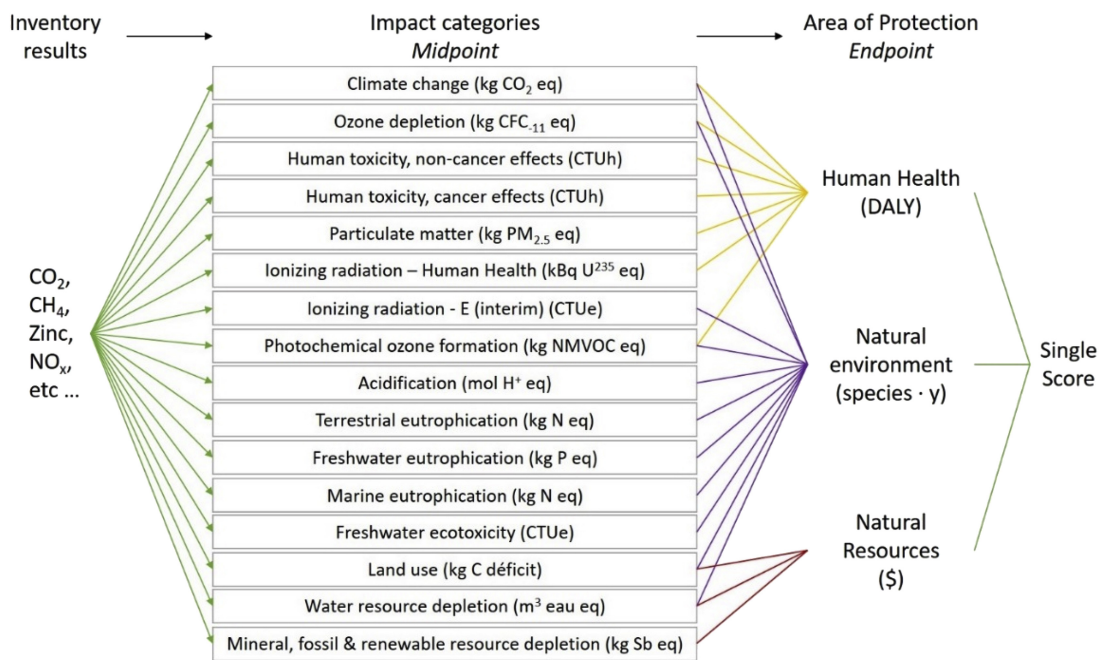


Figure 4.1: Method Ecological Footprint v3.0, [21]

4.2 Impact Contribution Analysis

In this section, we consider a large wind turbine design with a rotor diameter of 4.2 m and a mast height of 24 m. This configuration is commonly deployed, and its scale accentuates trends in the inventory, thereby facilitating the sensitivity analysis.

First, we identify the impact categories that contribute most significantly to the overall single score. As shown in Figure 4.2, the abiotic depletion of resources accounts for nearly 40% of the total score, followed by global warming and freshwater ecotoxicity.

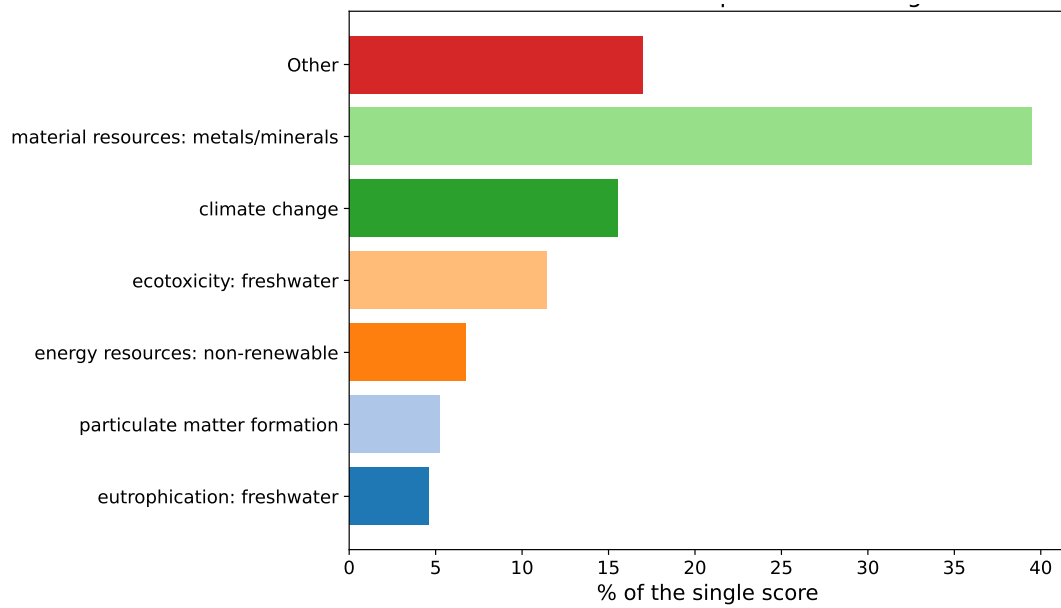


Figure 4.2: Relative contribution of each impact category to the single score. Only those categories contributing cumulatively up to 80% of the total score are shown.

Figure 4.3 further shows that nearly 80% of the abiotic depletion impact arises from copper used in the stator coils and the mast cable. In contrast, the global warming potential is primarily driven by the use of steel and concrete.

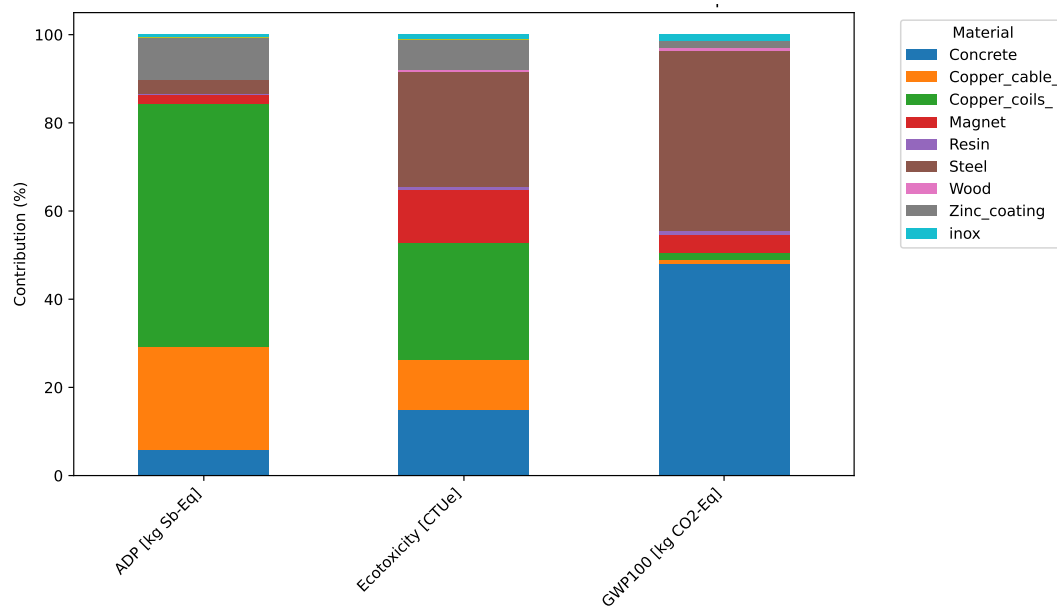


Figure 4.3: Relative contribution of key materials to the three largest impact categories.

Finally, Figure 4.4 highlights that the three materials contributing most to the single score are copper, concrete, and steel. When combining copper use in both the stator and the cables, it becomes the dominant contributor, largely due to its high impact on abiotic resource depletion. Interestingly, neodymium magnets—often flagged as ecologically problematic—rank only sixth, slightly behind zinc coating.

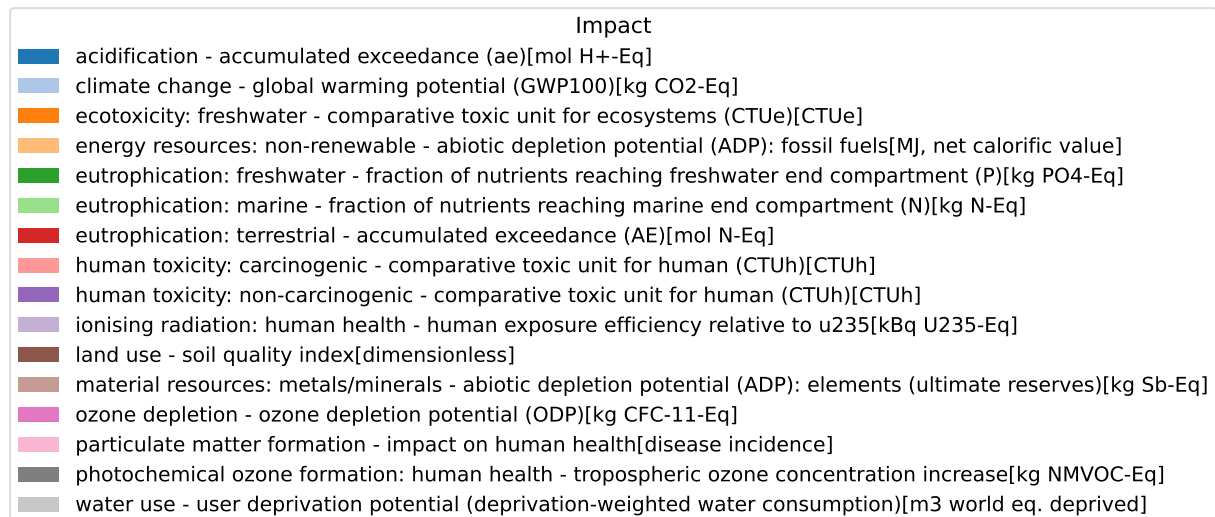
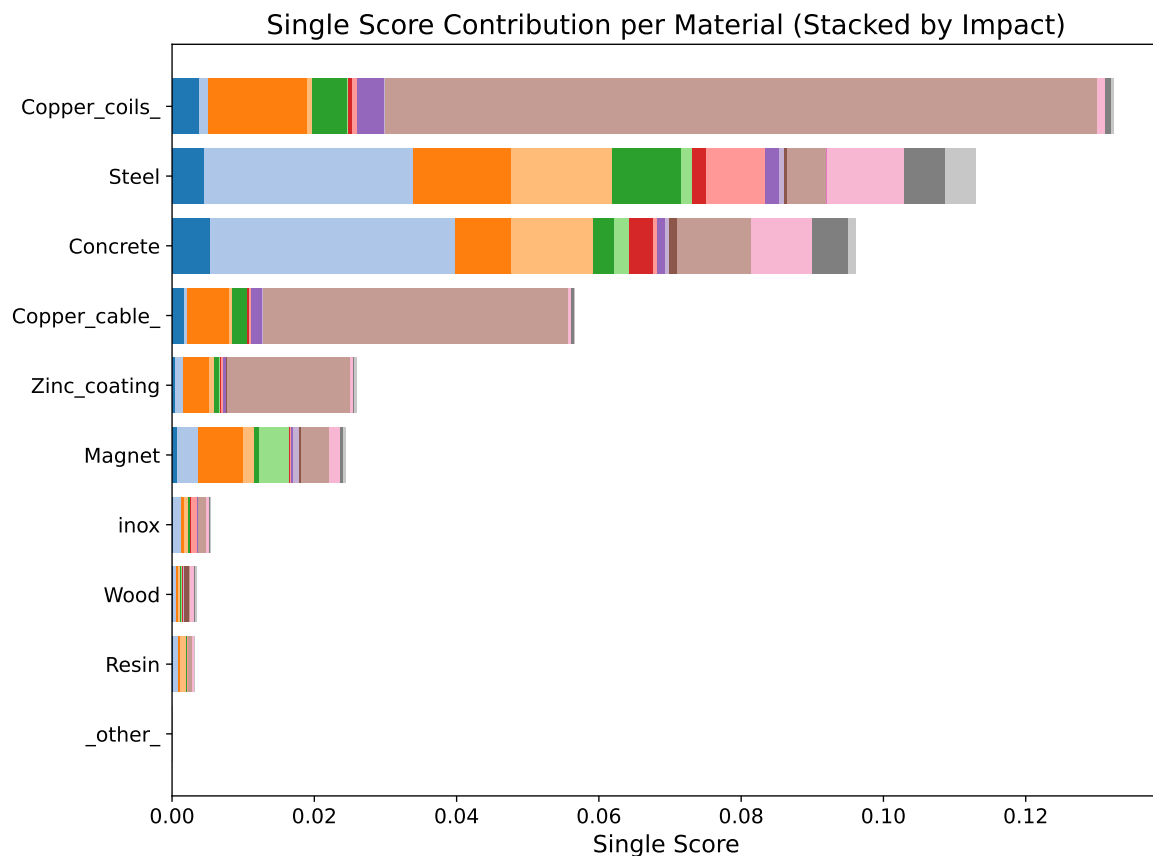


Figure 4.4: Relative contribution of each material to the single score, grouped by impact category.

The Figure 4.4 highlights the critical nature of copper. Although it represents only a tiny proportion of the mass of the total system, 8kg compared with around 300kg of steel or 4m³ of concrete, it is by far the most significant contributor to the single score.

However, these results should be interpreted with caution. They are strongly influenced by the methodological choice of using a single aggregated score, and the absence of uncertainty

analysis prevents us from treating the values as precise.

4.3 Sensitivity analysis

4.3.1 Absolute impacts

Table 4.1 shows the absolute impacts of the construction of a wind turbine for different sizes.

System	ADP(MR) [kg Sb-eq]	GWP [kg CO ₂ -eq]	ADP(Fossil) [MJ]	Eutroph. [kg PO ₄ -eq]	Ecotox. [CTUe]
Small system (h = 12 m, d = 1800 mm)	0.0620	1,093.81	46,642.7	0.503849	46,643
Medium system (h = 18 m, d = 3000 mm)	0.1018	1,641.76	14,986.3	0.8035	78,566.3
Large system (h = 24 m, d = 4200 mm)	0.1532	2,752.55	117,247	1.22089	117,247
Variation (Large / Small)	×2.47	×2.52	×2.51	×2.42	×2.51

Table 4.1: Main environmental impacts of three wind turbine configurations, with variation from the smallest to the largest.

To better understand how these impacts evolve with design choices, we examine their variation with both mast height and diameter.

Figure 4.5 shows that the impact on material resources varies slightly with mast height, +50% between 12m and 30m, whereas climate change more than doubled over the height range. This observation aligns with the results of the previous section. The increase in material resource use is primarily due to the electrical cable inside the mast, which scales linearly with height. In contrast, the climate change impact is driven by the use of steel and concrete, whose quantities grow approximately with the square of the mast height.

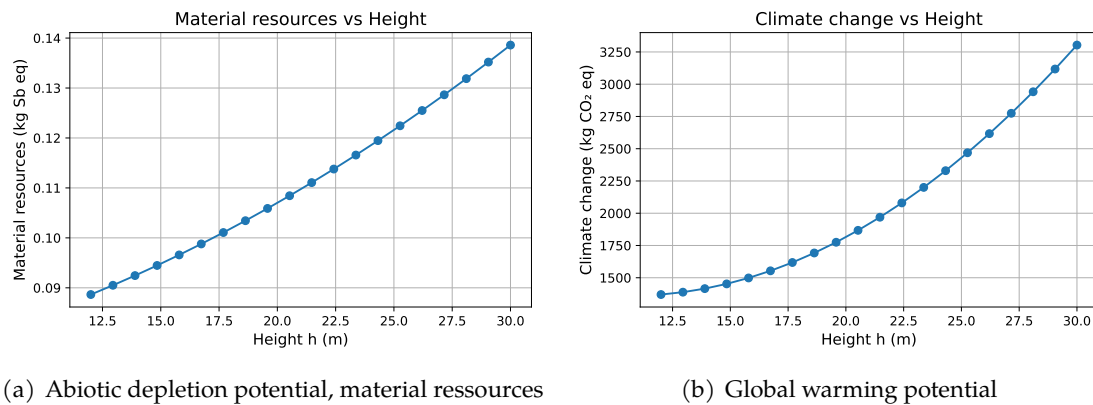


Figure 4.5: Variations of impacts with height, for a diameter of 3m

For the rotor diameter, the trends are almost reversed, as shown in Figure 4.6. Here, the climate change impact remains nearly constant, while the abiotic depletion of material resources approximately doubles. This pattern is consistent with the findings of the previous section. The increase in resource depletion is primarily due to the copper coils in the stator, which volume grows with the diameter. In contrast, the amount of steel in the generator remains relatively small, about ten times less than that used for the structure, and thus has a limited effect on the global warming potential.

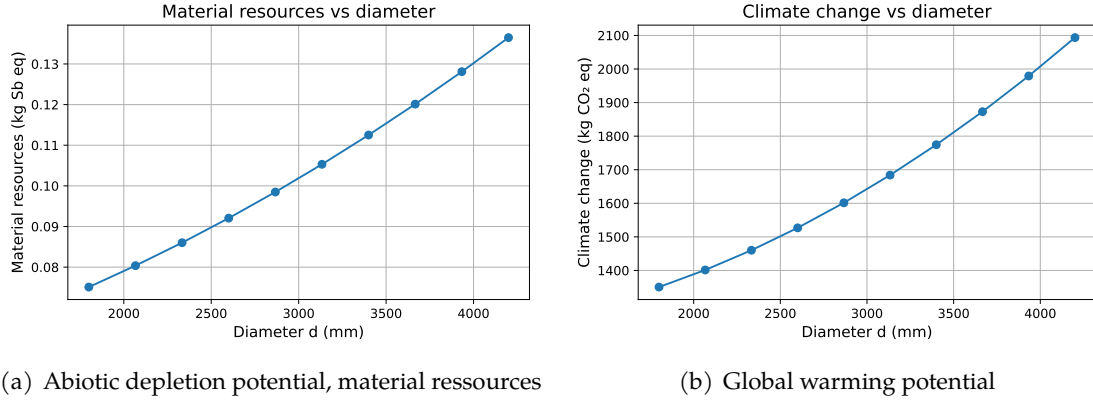


Figure 4.6: Variations of impacts with diameter, for a height of 18m

These results are summarized in the Sobol matrix (Table 4.2). Sobol indices, ranging from 0 to 1, quantify the proportion of output variance that can be attributed to each input parameter. A value close to 1 indicates a strong influence of the parameter on the output, while a value near 0 means negligible influence.

The matrix shows that the rotor diameter has very little effect on GWP, while it plays a dominant role in the ADP(MR), primarily due to its impact on copper usage. Conversely, mast height is the most influential factor for GWP due to its correlation with steel and concrete consumption.

	ADP(MR)	Ecotoxicity	Global Warming
h	0.40	0.65	0.87
d	0.60	0.35	0.13

Color scale:

- High sensitivity (> 0.8)
- Medium sensitivity ($\sim 0.5-0.7$)
- Low sensitivity ($\sim 0.3-0.4$)
- Minimal sensitivity (< 0.2)

Table 4.2: Sobol indices (share of variability) for height (h) and diameter (d) on key environmental impacts.

4.3.2 Impact per functional unit

The next step is to investigate the influence of design choices on impacts while considering the functional unit. For this subsection, the following characteristics have been chosen :

Parameter	Value	Description
Reference wind speed v_{ref}	5 m/s	Annual average wind speed at reference height
Reference height h_{ref}	12 m	Height at which v_{ref} is measured
Surface roughness z_0	0.5 m	Terrain roughness length (e.g., forested or built-up area)
Turbine lifetime	20 years	Expected operational lifespan of the wind turbine

Table 4.3: Wind site characteristics used for energy production assessment.

When considering the functional unit, an optimal mast height emerges with respect to environmental impacts, as shown in Figure 4.7. This optimum occurs around 20 m for the abiotic depletion potential (ADP(MR)) and approximately 15 m for the global warming potential (GWP). This difference can be explained by the sensitivity of each impact to the input parameters, as highlighted in Table 4.2.

Increasing the mast height leads to a rapid rise in GWP per unit of energy produced, due to the quadratic growth in steel and concrete use. In contrast, ADP(MR) is less sensitive to height,

and the gains in energy production from a taller mast outweigh the additional material resource use. Therefore, a higher mast remains beneficial for ADP(MR).

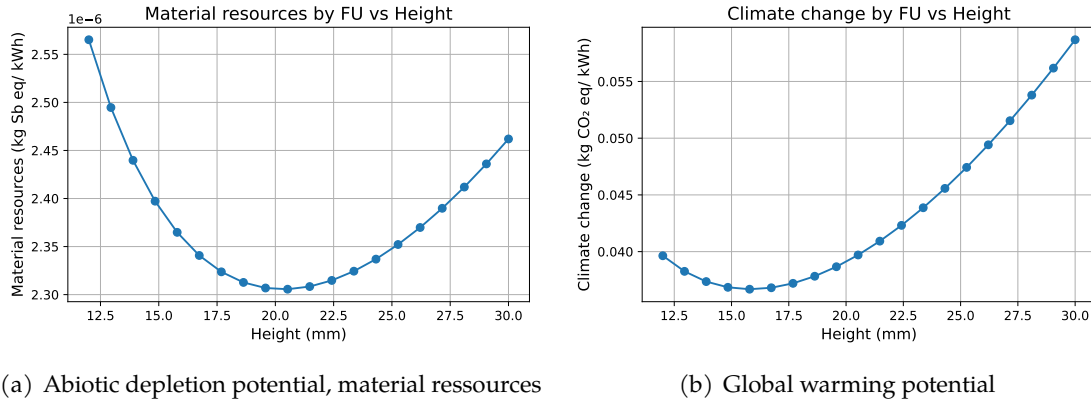


Figure 4.7: Variations of impacts by FU with height, for a diameter of 3m

Figure 4.8 shows that, in contrast to mast height, there is no trade-off when increasing the turbine diameter. For both impacts—ADP(MR) and GWP, the trend suggests that “the bigger, the better.” This can be explained by the fact the additional energy generated by a larger rotor more than offsets the resulting increase in material consumption.

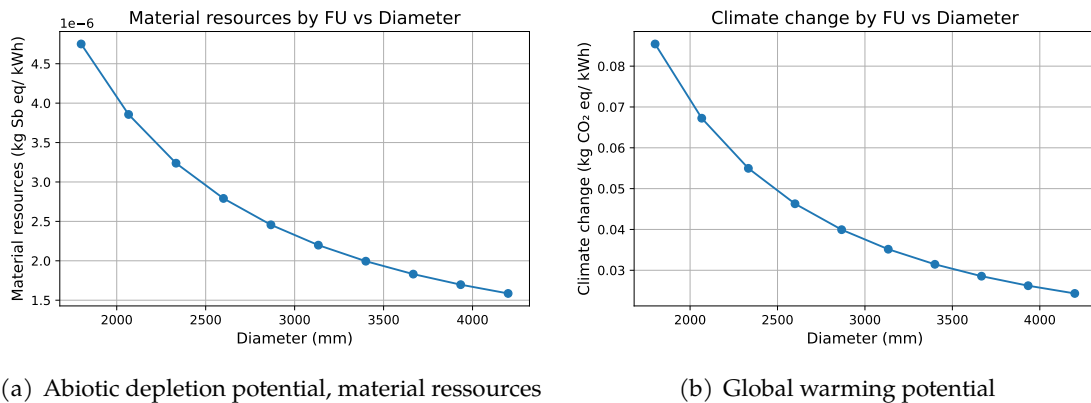


Figure 4.8: Variations of impacts per energy produced with diameter, for a height of 18m

4.4 Impact transfer

The findings of the previous sections highlight a trade-off between the two main design parameters: mast height and rotor diameter. Each parameter is the dominant driver of one of the two most significant impacts contributing to the single score. Specifically, mast height has the greatest influence on climate change, while rotor diameter predominantly affects the abiotic depletion of resources. This section aim to see if the optimisation of one impact could come at the cost of increasing the other.

To do so, four optimizations have been run. All of them aim to minimize a ratio between a given quantity and the energy production. These quantities are :

- The single score
- The abiotic depletion potential of minerals resources

- The global warming potential
- The sum of every masses of the inventory

The ratio between the single score and the total amount of energy produce is called **eco-efficiency**.

The inputs are :

- a geographical site, with it particular $v_{\text{ref}}, h_{\text{ref}}$ and z_0
- a lifetime
- an objective

We can thus define the following optimization problem (4.1).

$$\begin{aligned} \min_{h,d} \quad & \text{QtyEff}(z_0, v_{\text{ref}}, h, d) \quad \text{s.t.} \quad h_{\min} \leq h \leq h_{\max}, \\ & d_{\min} \leq d \leq d_{\max}, \end{aligned} \quad (4.1)$$

With

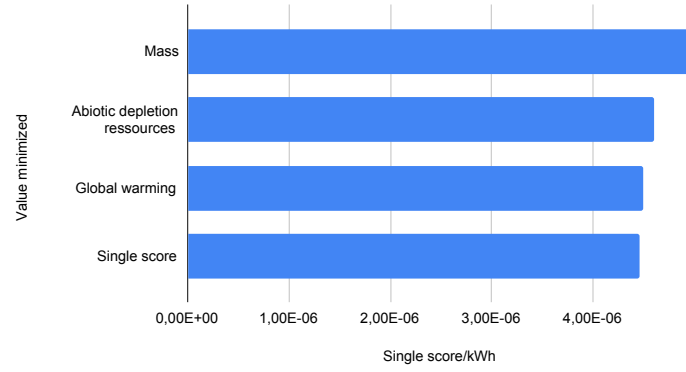
- QtyEff : Ration between a given quantity (single score, global warming potential etc.) and the total amount of energy produce.
- $h_{\min} = 12 \text{ m}, \quad h_{\max} = 30 \text{ m}$
- $d_{\min} = 1800 \text{ mm}, \quad d_{\max} = 4200 \text{ mm}$

The optimization returns a defined wind turbine design in the form of a height-diameter pair. Then, the impact of each configuration is calculated. All data are summarized, Table 4.4.

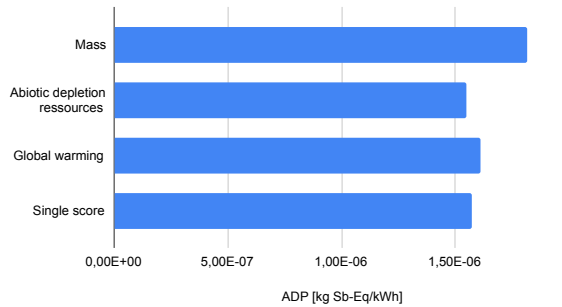
Optimization criterion	h [m]	d [mm]	Score [unit/kWh]	ADP(MR) [kg Sb-eq/kWh]	GWP [kg CO ₂ -eq/kWh]	Ecotox. [CTUe/kWh]
Mass	12.00	4200	4.98e-6	1.82e-6	2.67e-2	1.28
ADP(MR)	23.00	4200	4.60e-6	1.55e-6	2.70e-2	1.17
GWP	16.78	4200	4.49e-6	1.61e-6	2.42e-2	1.16
Single score	18.68	4200	4.45e-6	1.57e-6	2.45e-2	1.15

Table 4.4: Impacts per functional unit for different optimization criteria. Reference conditions: $h_{\text{ref}} = 12 \text{ m}, z_0 = 0.5, v_{\text{ref}} = 5 \text{ m/s}$.

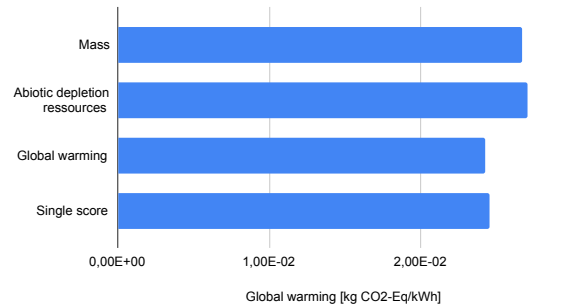
The results presented in Table 4.4 show that, for all optimization criteria, the diameter reaches the upper bound of the possible sizes. This confirms the trend observed in Figure 4.8, namely that there is no internal optimum with respect to diameter. In contrast, the trade-off lies in the mast height, where the optimum varies within the range of values depending on the optimization objective. For ADP(MR) and GWP, the results are similar to the trends shown in Figure 4.7. The optimization of the single score represents a compromise between these two impacts, while minimizing the total mass leads to a solution at the lower bound of the height range.



(a) Single score per energy produced for different configurations



(b) Abiotic depletion resources impact per energy produced for different configurations



(c) Global warming impact per energy produced for different configurations

Figure 4.9: Impacts per functional unit for various design parameters.

Figure 4.9 shows that each optimization seems to perform best in its respective target category. The single score optimization ranks second in both ADP(MR) and GWP, illustrating a balanced compromise. A trade-off between abiotic depletion potential (ADP(MR)) and global warming potential (GWP) is observed when comparing configurations optimized for one or the other. In Figure 4.9(b), the configuration that minimizes GWP per kWh results in a higher ADP(MR) impact. Conversely, in Figure 4.9(c), the configuration optimized for minimum ADP(MR) performs worst in terms of GWP. These results could highlight an impact transfer between the ADP(MR) and GWP categories.

Finally, mass optimization comes last or second last for each impact. This could indicate that simply trying to reduce the total mass of the system without looking at the specific impacts of each material results in configurations that are not optimal in terms of environmental impacts.

Once again, these results should be treated with caution. As the variations are fairly small, of the order of 10%, without taking uncertainties into account, it is not yet possible to say whether they are significant.

4.5 Optimum production set point

A classic engineering approach is to consider the need as an objective to be satisfied, while taking constraints into account. The aim of this section is to use the opposite logic and see the need as a variable that will enable the environmental footprint to be minimized.

To investigate this, a constraint was introduced into the optimization process. The objective now

becomes finding a pair of values (mast height and rotor diameter) that minimizes the single score per unit of energy produced, while ensuring a required minimum annual energy output. Six wind site configurations were studied, combining two reference wind speeds ($v_{\text{ref}} = 4$ or 5 m/s) and three surface roughness values ($z_0 = 0.1, 0.3$, or 0.5 m). This amounts to posing the optimization problem (4.2).

$$\begin{aligned} \min_{h,d} \quad & \text{EcoEff}(z_0, v_{\text{ref}}, h, d) \quad \text{s.t.} \quad h_{\min} \leq h \leq h_{\max}, \\ & d_{\min} \leq d \leq d_{\max}, \\ & FU(z_0, v_{\text{ref}}, h, d) = \text{NrjSP} \end{aligned} \quad (4.2)$$

With

- $h_{\min} = 12$ m, $h_{\max} = 30$ m
- $d_{\min} = 1800$ mm, $d_{\max} = 4200$ mm
- $FU(\dots)$: Functional unit i.e total amount of energy produced during the lifespan, in kWh
- NrjSP : Energy production set point, in kWh

The results are presented in Figure 4.10.

Note that the y-axis represents the total single score divided by the energy produced over 20 years, while the x-axis corresponds to the annual energy production. This representation facilitates interpretation from an energy planning perspective.

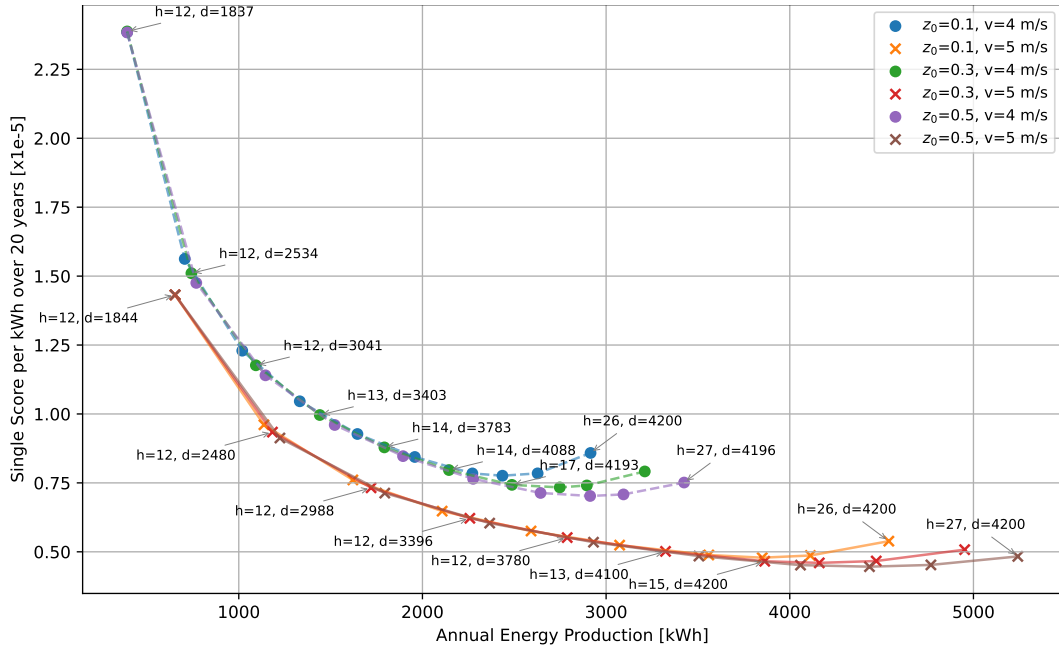


Figure 4.10: Single score per kWh as a function of annual energy production, for different roughness coefficients z_0 and reference wind speeds v_{ref} .

From the shape of the curves, it is clear that increasing either v_{ref} or z_0 raises the achievable energy production.

Moreover, the single score per unit of energy initially decreases with increasing production, reaches a minimum, and then slightly increases again. This suggests the existence of a production optimum that minimizes the environmental impact per kilowatt-hour.

Examining the evolution of the design parameters reveals that, at low production levels, the rotor diameter increases first, while the mast height remains relatively constant. Only at higher production requirements does the mast height begin to rise. These evolutions are shown in Figure 4.11.

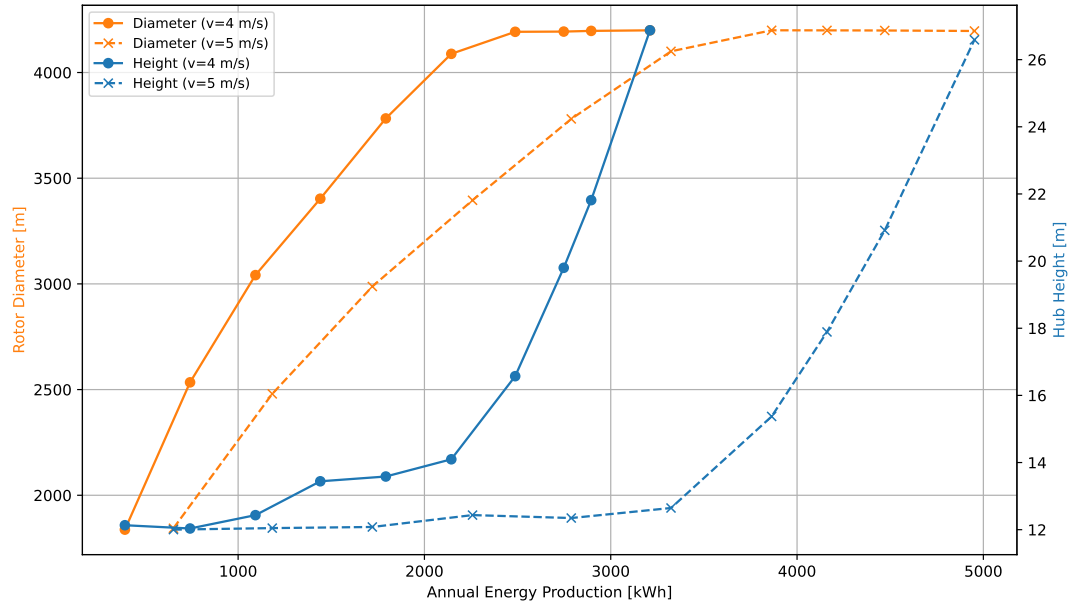


Figure 4.11: Evolution of the height and the diameter as a function of energy production for $z_0 = 0.3\text{m}$ and different reference wind speeds v_{ref} .

The configurations with the lowest environmental impact per kWh consistently feature the largest diameters. This indicates that when energy needs do not justify a large rotor, “environmental performance” deteriorates.

However, it is also essential to consider the absolute impacts. Figure 4.12 presents the absolute single score for each configuration obtained through the constrained optimizations.

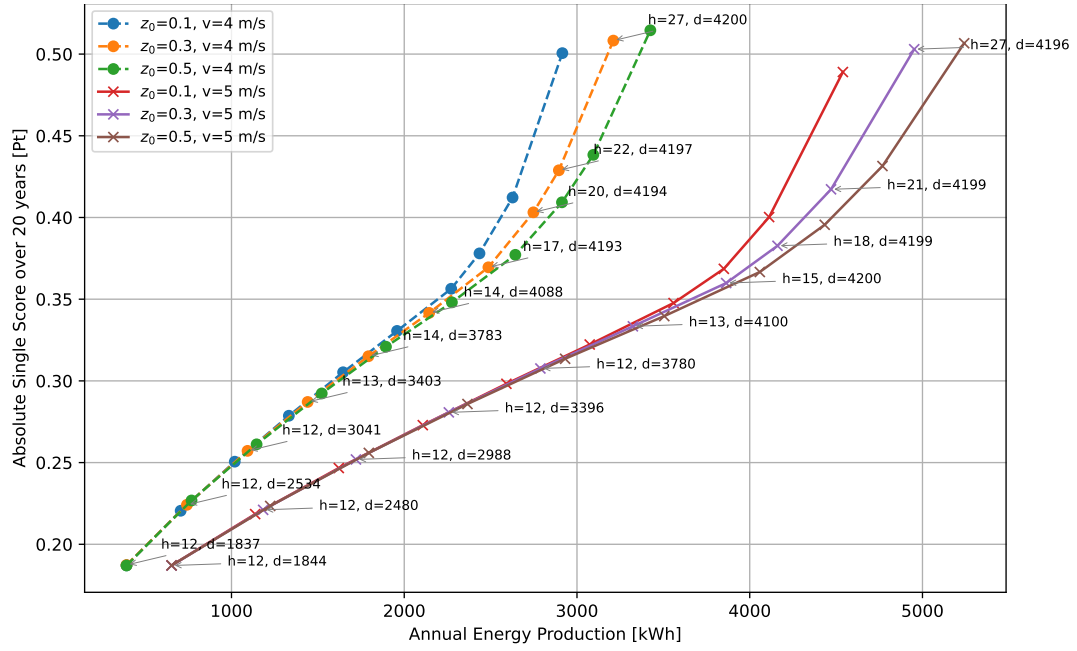


Figure 4.12: Absolute single score as a function of annual energy production, for different surface roughness values z_0 and reference wind speeds v_{ref} .

A noticeable breakpoints appears in each curve, corresponding to the point where the rotor diameter reaches its upper limit. Beyond this threshold, any further increase in energy production can only be achieved by raising the mast height. This results in a sharp rise in the single score, which grows almost exponentially.

So, according to the results found here and within the framework of the hypotheses, from an environmental point of view a large wind turbine on a small mast seems more interesting than a small wind turbine on a large mast.

The main objective of this work was to explore design trade-offs between mast height and rotor diameter, in order to support the locally manufactured wind turbine community in selecting the most sustainable system configuration based on specific installation site conditions. To achieve this, a Python-based tool was developed to perform parametric life cycle assessments using the Brightway 2 framework. This tool integrates outputs from the OpenAFPM software to build a parametric inventory, and uses the NoLoad optimization library to identify optimal designs. Background data were sourced from the ecoinvent 3.8 cut-off database, and results were analyzed using the Product Environmental Footprint method.

This study showed that :

- The impact categories that contribute most to the single score are the abiotic depletion potential of mineral and metal resources, the global warming potential and the eco toxicity.
- The main contributor to ADP(MR) is the copper. For the GWP it is the steel and the concrete.
- The height of the wind turbine therefore has the greatest influence on GWP, whereas ADP(MR) is mainly driven by the diameter of the wind turbine.
- Finding a height-diameter configuration that minimizes the ratio between an impact and the amount of energy produced is to the detriment of the other.
- Optimizing the single score allows a compromise between the two. Optimizing only the mass of the system does not seem relevant.
- There is an optimum for mast height, but no internal optimum for rotor diameter within the considered range.
- There is an optimum for energy production. Low energy productions which do not justify a large diameter tend to deteriorate the “environmental performance”.
- From an environmental point of view, a large wind turbine on a small mast seems more interesting than a small wind turbine on a large mast.

All these results must be taken with caution. End-of-life considerations and uncertainties need to be taken into account before any recommendations can be drawn up.

Avenues for future work include :

- Goal and scope definitions :
 - Implementation of end-of-life data in the LCA
 - Broaden the scope to include electronics and batteries in particular.
- Inventory :
 - Update the database to eco-invent 3.11 and compare.
 - Use step regressions rather than smooth ones, especially for the guy wires.
- Tools :
 - Fix the compatibility problem between NoLoad and Brightway 2 so that faster, derivative-based optimization methods can be used.
 - Fix the compatibility problem between lca.algebraic and Brightway 2 for the calculation of uncertainties.

- Impact assessment :
 - Use another method than PEF to confirm the importance of ADP(MR) and therefore copper in the single score.
 - Investigate different EoL scenarios, especially for copper.
 - Investigate more geographical sites, with more different z_o and v_{ref} .

6.1 Appendix 1 : Complete inventory

	Sub-system	Element	Element in french	Part	Process (Ecovinvent 3.8)	Lifetime/End of life	Comments
Wind generator	Propeller	Blades	Pales	Dry soft wood	planing, beam, softwood, u=20% -Europe without Switzerland	With proper annual maintenance and protective tape on the leading edge, the rotor blade is a wear part but can last up to 20 years. A common issue is damage caused by the tail vane or leading-edge erosion due to wind, rain, or hail. Repairs are possible using epoxy resin, which provides good durability. At end of life, the blade can be incinerated or disposed of with wood waste.	Disc and triangle that hold the blades in place
		Blades attachment	Fixation des pales	Plywood	plywood production-RER		
	Stator	Copper coil	Bobine de cuivre	Copper, cathod	market for copper, cathode-GLO	Copper recovery is sometimes accepted by metal recyclers, although not always. At end of life, the resin can be burned or melted, allowing copper to be recovered. Without protective paint, the resin degrades under UV exposure and rain, leaving the fiberglass exposed—resulting in a lifespan of around 5 years. If protected, the coating should be renewed every 5 to 10 years, enabling a service life of 20 years or more. Main failure risks include short circuits, impacts, or excessive friction. In most cases, the stator is not recovered, as it tends to break more easily than other components.	
				Sheet rolling	sheet rolling, copper-RER		
				Wire drawing	wire drawing, copper-RER		
		Enamelling	Revêtement isolant	Resin	polyester resin production, unsaturated,RER		
				Xylene	xylene production,RER		
				Electricity	market group for electricity, medium voltage,RER		
		Resin	Résine	Polyester resin	market for polyester resin, unsaturated		
				Talc	steatite quarry operation-RoW		
				Glass fibre	glass fibre production-RER		
	Rotor	Metal disk	Disque de métal	Steel, low-alloyed	permanent magnet production, for electric motor-RER	Corrosion issues may arise, particularly for neodymium components due to their alloy composition. To prevent this, proper painting and galvanized discs are used. With adequate surface protection, the part can last up to 20 years. It is generally recoverable due to its mechanical strength and is, along with the stator, one of the most valuable components.	
				Zinc coat	zinc coating, pieces-RER		
		Magnet	Aimant	permanent magnet for motor	permanent magnet production, for electric motor-GLO		
		Resin	Résine	Polyester resin	market for polyester resin, unsaturated		
	Hub			Talc	steatite quarry operation-RoW	This component is typically recovered from vehicles. If the seal is not watertight, water and particles can infiltrate. When properly sealed, only disassembly and re-greasing are needed, and the part can last up to 20 years.	
				Glass fibre	glass fibre production-RER		
	Tail	Vane	Plaque du safran	Plywood	plywood production-RER	Steel is not recommended due to vibrations that can damage the mounting plate. Low-quality plywood has a lifespan of around 10 years, while high-quality plywood can last up to 20 years. A condition check every 5 years is advised to determine whether repainting is needed. If properly treated from the start, repainting may only be required once during the turbine's lifetime. The part is easy to replace. If broken, it can be incinerated or disposed of with wood waste.	
		Boom pipe	Tube du safran	Steel, low-alloyed	steel production, converter, unalloyed-RER		
				Drawing of pipe	drawing of pipe, steel-RER	Occasional welding defects may occur, but otherwise the component has a service life of up to 20 years.	
		Hinge outer	Charnière extérieur	Steel, low-alloyed	steel production, low-alloyed, hot rolled-RER		
				Drawing of pipe	drawing of pipe, steel-RER		
		Bracket , support and tail stop	Equerre, renfort et encocge d'arrêt	Steel, low-alloyed	steel production, low-alloyed, hot rolled-RER		
		Nacelle	Yaw bearing	Steel, low-alloyed	steel production, low-alloyed, hot rolled-RER	Can last up to 20 years if properly manufactured. However, wear can occur at the pivot point.	
				Drawing of pipe	drawing of pipe, steel-RER		
		Frame	Berceau	Steel, low-alloyed	steel production, low-alloyed, hot rolled-RER		
	Hardware	Stainless steel hardware	Boulonnerie inox	Stainless steel	steel production, chromium steel 18/8, hot rolled-RER	No specific issues reported. Has a service life of 20 years but may deform in the event of a fall.	
				Average work	metal working, average for chromium steel product manufacturing-RER		
		Zinc coated hardware	Boulonnerie acier zingué	Steel, low-alloyed	steel production, low-alloyed, hot rolled-RER	20 years service life	
				Average work	metal working, average for steel product manufacturing-RER		
	Other elements	Moulds	Moule	Plywood	zinc coating, pieces-RER	One mold can be used to produce at least five wind turbines. If insufficiently greased, wood may be torn during demolding.	
				Acetone			
		Coil winder	Enrouleur de bobine	Plywood		10 years of service life	

System	Sub-system	Element	Element in french	Part	Comments	Lifetime / End of life	Size	Attribute	Formula	Value	Unit	Process (Ecovinvent 3.8)
Structure	Foundation	Concrete	Béton		Foundation for the mast and the guy wires. The quantity of concrete depended on the height of the mast and the diameter of the wind turbine.	Technically, the foundation can be removed. The concrete block can be crushed, and the reinforcing bars recovered. However, in most cases, it is left in the ground. It is also possible to remove only the upper part and leave the rest buried. The foundation can easily last over 100 years and may be reused if the wind turbine is replaced.					kg	concrete block production-RoW
		Concrete reinforcing bar HA10	Fer à béton HA10	Reinforcing steel	Reinforcing bar use to make reinforced concrete. The linear mass come from the european standards		L = f(h) D = 10mm	Linearar mass : 0,617 kg/m	L * linear mass		f(h) kg	reinforcing steel production
		Anchor	Ancrage	Reinforcing steel	Reinforcing bar galvanized use to make the anchor for the guy wires. The linear mass come from the european standards		L = 8,5m D = 16mm	Linearar mass : 1,578 kg/m	L * linear mass	13,4	kg	reinforcing steel production
				Zinc coat					$\pi^8,5^*0,016$	0,427	m²	zinc coating, pieces-RER
	Mast of height h	Mast pipe	Tube du mât	Steel, unalloyed	Steel pipes use to build the mast. Several diameters are possible. depending of the height of the mast.	Sometimes galvanized. Has a very long service life and is rarely replaced. Reuse is possible, otherwise, it can be sent to a metal recycler.	D1 = 88,9 mm D2 = 114,3 mm e = 3,2 mm L1 = f(h) L2 = f(h)	$\rho = 7850 \text{ kg/m}^3$	$\rho^* \pi^8,5^* ((D1-e)^*L1 + (D2-e)^*L2)$	f(h)	kg	steel production, converter, unalloyed-RER
				Drawing of pipe	//		//	//	//	//	//	drawing of pipe, steel-RER
		Yaw pipe	Flèche de manoeuvre	Steel, unalloyed	The yaw pipe helps raising the tower when it is lowered. Its diameter depends on the size of the mast and its length is half that of the mast		e = 3,2 or 2,9 mm D = 88,9 or 60,3 mm	$\rho = 7850 \text{ kg/m}^3$	$\rho^* \pi^8,5^* e^2 (D-e)^2$	f(h)	kg	steel production, converter, unalloyed-RER
				Drawing of pipe	//		//	//	//	//	//	drawing of pipe, steel-RER
		Mast pivot tube	Tube pivot du mât	Steel, unalloyed	The tube that joins the mast to the wind turbine. Its diameter and it legnt depends on the height of the mast		e = 6mm D = 60,3 or 76 mm L = 395 or 480 mm	$\rho = 7850 \text{ kg/m}^3$	$\rho^* \pi^8,5^* (D-e)^2$	f(h)	kg	steel production, converter, unalloyed-RER
				Drawing of pipe			//	//	//	//	//	drawing of pipe, steel-RER
		Steel swivel pin	Axe de pivot	Steel, low-alloyed	Mast pivot axis. A constant average value, M18, has been taken for the size of the threaded rod.		M18*1000 mm L=1 m D=0,018m	$\rho = 7850 \text{ kg/m}^3$	$\rho^* \pi^8,5^* D^2 * L/4$	2,00	kg	steel production, low-alloyed, hot rolled-RER
				Average work			//	//	//	//	//	metal working, average for steel product
				Zinc coat					$\pi^8,5^* D^2 * L$	0,057	m²	zinc coating, coils-RER
		Gusset	Gousset	Flat iron	Gusset plates welded to the mast. These are used to attach the shrouds to the mast.		50*8 mm L = 0,113*h + 0,68	$\rho = 7850 \text{ kg/m}^3$	$\rho^* L^2 * 0,05^* 0,008$	f(h)	kg	steel production, low-alloyed, hot rolled-RER
		Base	Embase	Flat iron	Used to secure the mast to the concrete block		100*10 mm L = 1m	$\rho = 7850 \text{ kg/m}^3$	$\rho^* L^2 * 0,1^* 0,01$	7,85	kg	steel production, low-alloyed, hot rolled-RER
				Iron corner			60*60*8 mm L = 1m	$\rho = 7850 \text{ kg/m}^3$	$\rho^* L^2 * 0,06^* 0,008^2$	7,536	kg	steel production, low-alloyed, hot rolled-RER
		Welding	Soudure		The total length of the weld is highly variable. The value used corresponds to the use of a whole pack of welding rods, i.e. 25 rods of 30 cm.		5 m					welding, arc, steel-RER
	Rigging	Zinc coated hardware	Quincaillerie zinguée	Steel, low-alloyed	All the screws, nuts, cable clamp etc. used to build the structure. Most of the weight come from the hardware for the rigging. Pieces are galvanized.	Hardware components are difficult to reuse. Regular inspection of tightening is recommended. They can last up to 20 years, with replacement carried out if necessary.			$0,92^* h + 1,28$	f(h)	kg	steel production, low-alloyed, hot rolled-RER
				Average work					//	//	kg	metal working, average for steel product
				Zinc coat					$0,0552^* h + 0,0771$	f(h)	m²	zinc coating, pieces-RER
		Guy wire	Hauban	Steel, unalloyed	The shrouds keep the mast upright. Each layer consists of four shrouds, located at the front, back and sides of the turbine. The number of layers depends on the height of the mast. We assume a constant diameter but it varies according to the manual.	Very long service life. However, in marine environments, it may be shorter. Rust can sometimes cause it to seize, requiring replacement after approximately 15 years. It is recoverable, and if broken, it can be sent to a metal recycler.	D = 6 mm L = 38,6 + -2,56*h + 0,564*h^2	$\rho = 7850 \text{ kg/m}^3$ k = 0,51 (fill factor)	$k^* \rho^* \pi^8,5^* D^4 * L$	f(h)	kg	steel production, converter, unalloyed-RER
				Wire drawing	//		//	//	//	//	//	wire drawing, steel-RER
				Zinc coat	The guy wires are coated with zinc to protect them.				$D^2 * \pi^8,5^* L$	f(h)	m²	zinc coating, pieces-RER
		Guy wire distributor	Répartiteur haubans	Flat iron	Used to attach guy wires to anchors on concrete blocks		150*6 mm L = 1200 mm	Linear mass = 9,42 kg/m	L * Linear mass	11,304	kg	steel production, low-alloyed, hot rolled-RER
	Cable 48V	Copper wire	Câble en cuivre	Copper cathode	Cable between generator and rectifier	The cable can become twisted. If it is not unplugged and untwisted regularly, excessive twisting may occur, potentially causing a short circuit and damaging the stator. With proper maintenance, the cable does not need to be replaced and can last up to 20 years.	S = 4 mm² n = 3 (Three-phase)	$\rho = 8960 \text{ kg/m}^3$	$\rho^* S^2 * h^2 * n$	f(h)	kg	market for copper, cathode-GLO
				Wire drawing	//		//	//	//	//	//	wire drawing, copper-RER
		Plastic	Plastique	Polyethylene	Electrical insulation		D = 11,9 mm	$\rho = 950 \text{ kg/m}^3$				polyethylene production, high density, granulate-RER
				Extrusion plastic			//	//	//	//	//	extrusion, plastic pipes-RER
				Machine						2,12E-11	unit	rolling mill production-RER
				Tap water						17,424*h^12	kg	market for tap water-Europe without Switzerland

System	Sub-system	Element	Element in french	Part	Lifetime/End of life	Process (Ecovincent 3.8)	
Electronics	Battery	Lead acid Battery	Batterie plomb	Battery, lead acid, re	Lead-acid batteries have a lifespan of 5 to 10 years. There is limited feedback on lithium-based batteries due to their relatively recent adoption. At end of life, both types can be sent to a metal recycler.	market for battery, lead acid, rechargeable, stationary-GLO	
	Resistor	Dump load	Résistance de décha	Resistor, auxiliaries	Recyclé au fer. 20 ans	resistor production, auxiliaries and energy use-GLO	
					A common issue is voltage spikes, particularly due to lightning. Capacitors have a lifespan of 10 to 15 years. As they are difficult to replace, the entire component is often discarded. However, it is possible to dismantle certain parts, such as the heat sink, and send the rest to a metal recycler.		
	Inverter	Inverter 500 W	Onduleur	Inverter, 0.5kW		inverter production, 0.5kW-RER	
	Charge controller (OSWACC)	Components	Composants	Capacitors		market for capacitor, auxiliaries and energy use	
				Resistors, metal film		market for resistor, metal film type, through-hole mounting	
				Resistors, wirewound		market for resistor, wirewound, through-hole mounting	
				Potentiometer		market for potentiometer, unspecified	
				Diodes		market for diode, glass-, for through-hole mounting	
				Transistors		market for transistor, auxiliaries and energy use	
				Integrated circuits		market for integrated circuit, logic type	
		PCB	PCB	PCB		market for printed wiring board, through-hole mounted, unspecified, Pb free	
				Soldering Tin		tin production, RoW	
		Heatsink	Dissipateur thermiqu	Aluminium for heatsink		market for aluminium, primary, ingot	
				Extrusion of aluminium		market for impact extrusion of aluminium, 1 stroke	
				Threading of heatsink		market for metal working, average for aluminium product manufacturing	
				Anodization of heatsink		market for anodising, aluminium sheet	
		Mounting clips	Fixation	Steel for transistor mounting clips		market for steel, low-alloyed, hot rolled	
				Shaping of transistor mounting clips		market for metal working, average for steel product manufacturing	
		Case	Boîte	ABS case		market for acrylonitrile-butadiene-styrene copolymer	
		Connectors	Connecteurs	Connectors		market for electric connector, peripheral type buss	
				Switch		market for switch, toggle type	
				Temp. Sensor cable		cable production, connector for computer, without plugs	
	Rectifier	Rectifier	Redresseur	Diode, glass-, for sur	Rarely requires replacement and has an estimated service life of 20 years. At end of life, it can be sent to a metal recycler.	diode production, glass-, for surface-mounting-GLO	
				Copper		market for copper, cathode-GLO	
				Polycarbonate		market for polycarbonate-GLO	
		Rectifier heatsink	Dissipateur thermiqu	Aluminium, primary, i	Can be recovered from an inverter. Recyclable, can be disposed of in an aluminum recycling bin.	market for aluminium, primary, ingot-IAI Area, EU27 & EFTA	
				Anodization of heatsink		market for anodising, aluminium sheet	
				Extrusion of aluminium		market for impact extrusion of aluminium, 1 stroke	
	Cables	Copper wire	Câble en cuivre	cable, three-conductor	Can be sent to a metal recycler, either bare or with insulation. If no issues arise, it does not require replacement and can last up to 20 years.	market for cable, unspecified	
				Cable, one-conductor		market for cable, unspecified	
	Circuit breaker		Disjoncteur	three poles circuit bre	Easily last 20 years. Can be sent to a metal recycler.	market for switch, toggle type	
	Transport	Container	Contenaire			market for transport, freight, sea, container ship-GLO	
		Barge	Péniche			market for transport, freight, inland waterways, barge tanker-RER	
		Train	Train			market group for transport, freight train-GLO	
		Lorry	Camion			market group for transport, freight, lorry, unspecified-GLO	
		Light commercial veh	Camionnette			market group for transport, freight, light commercial vehicle-GLO	

6.2 Appendix 2 : Computation of the mass of hardware for generator and structure

6.2.1 Hardware for the structure

Mast Height [m]	Gusset [m]	Expansion Bolts[kg]	Nuts [kg]	Turnbuckles [kg]	Shackles [kg]	Cable Clamps [kg]	Thimbles [kg]
12	2.00	0.144	0.1998	5.05	2.40	4.32	0.08
18	2.80	0.348	0.3864	6.65	4.31	6.24	0.168
24	3.36	0.348	0.3864	9.05	4.56	8.64	0.248

Table 6.1: Mass of individual hardware components as a function of mast height.

Height [m]	Zinc hardware [kg]	Coated Area [m ²]
12	12.1938	0.7316
18	18.1024	1.0861
24	23.2324	1.3939

Table 6.2: Total zinc hardware for structure mass and corresponding coated surface area.

6.2.2 Hardware for the generator

Rotor Diameter	1.80 m	2.40 m	3.00 m	3.60 m	4.20 m
Stainless Steel Components					
Threaded rods	M12×1500	M12×1500	M12×1500	M14×2000	M14×2000
Mass [kg]	1.33	1.33	1.33	2.42	2.42
Nuts	48	48	48	60	60
Mass [kg]	0.83	0.83	0.83	1.50	1.50
Washers	12	12	12	23	23
Mass [kg]	0.12	0.12	0.12	0.23	0.23
Screws	Ø5×40mm	Ø5×50mm	Ø5×50mm	Ø5×60mm	Ø5×75mm
Quantity	42	54	66	66	78
Mass [kg]	0.20	0.31	0.38	0.45	0.65
Subtotal stainless [kg]	2.48	2.59	2.66	4.59	4.80
Zinc-Coated Components					
Axle bolt	M12×40mm	M12×40mm	M12×40mm	M14×40mm	M14×40mm
Mass [kg]	0.14	0.14	0.14	0.14	0.14
Tail bolt	M6×30mm	M6×30mm	M6×30mm	M6×30mm	M6×30mm
Mass [kg]	0.03	0.03	0.03	0.03	0.03
Screws 5×50mm	1 box	1 box	1 box	1 box	1 box
Mass [kg]	0.575	0.575	0.575	0.575	0.575
Screws 5×30mm	1 box	1 box	1 box	1 box	1 box
Mass [kg]	0.282	0.282	0.282	0.282	0.282
Threaded rods (zinc)	M10×1000 + M12×1000				
Mass [kg]	1.50	1.50	1.50	1.50	1.50
Nuts (zinc)	10×M10	10×M10	10×M10	10×M10 + 3×M12	same
Mass [kg]	0.116	0.116	0.116	0.168	0.168
Washers (zinc)	2×M10	2×M10	2×M10	2×M10	2×M10
Mass [kg]	0.02	0.02	0.02	0.02	0.02
Subtotal zinc [kg]	2.67	2.67	2.67	2.72	2.72

Table 6.3: Mass and quantity of hardware components for different rotor diameters

Rotor Diameter [m]	Stainless Steel Mass [kg]	Zinc-Coated Mass [kg]
1.80	2.48	2.67
2.40	2.59	2.67
3.00	2.66	2.67
3.60	4.59	2.72
4.20	4.80	2.72

Table 6.4: Total mass of stainless and zinc-coated hardware by rotor diameter.

6.3 Appendix 3 : Inputs for OpenAFPM CAD

Diameter	1m80	2m40	3m	3m60	4m20	Reference
Furling						
Vertical Plane Angle [°]	20	20	20	20	20	p.33
Horizontal Plane Angle [°]	55	55	55	55	55	p.33
Bracket Length [mm]	300	300	400	500	600	Table 10 p.36
Bracket Width [mm]	30	30	30	30	30	Table 10 p.37
Bracket Thickness [mm]	3	3	3	3	3	Table 10 p.38
Boom Length [mm]	800	1000	1200	1500	1800	Table 9 p.35
Boom Pipe Diameter [mm]	48.3	48.3	48.3	48.3	48.3	Table 9 p.35
Boom Pipe Thickness [mm]	3.2	3.2	3.2	3.2	3.2	Standard thickness for this diameter
Vane Thickness [mm]	6	10	10	12	12	Table 10 p.36
Vane Length [mm]	1000	1200	1300	1700	2000	Table 10 p.36
Vane Width [mm]	510	635	890	890	1145	Table 10 p.36
Offset [mm]	100	125	150	200	250	Table 4 p.30, Table 6 p.32
User						
Blade Width [mm]	124	124	124	124	124	?
Hub Pitch Circle Diameter [mm]	100	100	100	118.5	118.5	p.106
Rotor Disk Central Hole [mm]	76	76	76	86	86	p.106
Error [%]	1.40	1.03	0.74	–	–	
Holes Diameter [mm]	12	12	12	14	14	Table 11 p.54
Metal Length L [mm]	50	50	50	50	50	Table 3 p.29, Table 5 p.31
Metal Thickness L [mm]	5	5	5	5	5	Table 3 p.29, Table 5 p.31
Flat Metal Thickness [mm]	10	10	10	10	10	Table 6 p.32
Yaw Pipe Diameter [mm]	60.3	60.3	60.3	88.9	88.9	Table 9 p.35
Pipe Thickness [mm]	5	5	5	5	5	Below Table 9 p.35
Rotor Resin Margin [mm]	5	5	5	5	5	Below Table 3 p.42
Hub Holes Diameter [mm]	12	12	12	14	14	p.106

Table 6.5: Geometric and mechanical parameters used as inputs for OpenAFPM CAD, with references from the Tripalium handbook (2023).

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